

MS-GARCH Likelihood Approximation Study

Complete Model Parameter Reference

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Companion document to: *Impact of Likelihood Approximation Choices on the Performance of MS-GARCH Models*

June 2026

About this document. This file contains the complete estimated parameter sets for all 220 MS-GARCH(1,1) model runs reported in the companion article. Models are organized by market archetype (Section 1–5), then by asset, and within each asset by specification (Baseline and Exogenous) and approximation method (Gray’s collapsing, Plug-in, Gaussian Quadrature, Particle Filter). Parameter tables present the twelve-element vector $\Psi = \{c_1, c_2, \gamma_1, \gamma_2, \omega_1, \alpha_1, \beta_1, \omega_2, \alpha_2, \beta_2, p_{11}, p_{22}\}$ alongside the log-likelihood (LogLik) and convergence status (Conv.). Equation blocks then render the fitted model explicitly for each method. Dashes (–) in the γ columns indicate the baseline (internal-dynamics-only) specification where $\gamma_1 = \gamma_2 = 0$ by construction.

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1. Equity Market

Pivot (exogenous conditioning variable): GSPC.

1.1. Asset: FTSE

Baseline Specification

Table 1: FTSE — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0720	-0.0001	–	–	0.2687	0.3661	0.6337	0.0001	0.0001	0.0001	0.7607	0.4448	Yes	-1014.99
Plug-in	-0.0003	0.0355	–	–	0.0001	0.0001	0.0001	0.1062	0.2989	0.6954	0.5689	0.7553	Yes	-753.01
Quadrature	0.0517	-0.0001	–	–	0.2244	0.3836	0.6159	0.0001	0.0001	0.0001	0.7258	0.4785	Yes	-1044.07
Particle Filter	0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3030.64

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0720 \quad (1)$$

$$\mu_2 = -0.0001 \quad (2)$$

$$\sigma_t^2(1) = 0.2687 + 0.3661 y_{t-1}^2 + 0.6337 \sigma_{t-1}^2 \quad (3)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (4)$$

$$P = \begin{pmatrix} 0.7607 & 0.2393 \\ 0.5552 & 0.4448 \end{pmatrix} \quad (5)$$

LogLik = -1014.99; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0003 \quad (6)$$

$$\mu_2 = 0.0355 \quad (7)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (8)$$

$$\sigma_t^2(2) = 0.1062 + 0.2989 y_{t-1}^2 + 0.6954 \sigma_{t-1}^2 \quad (9)$$

$$P = \begin{pmatrix} 0.5689 & 0.4311 \\ 0.2447 & 0.7553 \end{pmatrix} \quad (10)$$

LogLik = -753.01; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0517 \quad (11)$$

$$\mu_2 = -0.0001 \quad (12)$$

$$\sigma_t^2(1) = 0.2244 + 0.3836y_{t-1}^2 + 0.6159\sigma_{t-1}^2 \quad (13)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (14)$$

$$P = \begin{pmatrix} 0.7258 & 0.2742 \\ 0.5215 & 0.4785 \end{pmatrix} \quad (15)$$

LogLik = -1044.07; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (16)$$

$$\mu_2 = 0.0000 \quad (17)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (18)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (19)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (20)$$

LogLik = -3030.64; Converged: Yes.

Exogenous Specification**Table 2:** FTSE — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0213	-0.0001	0.2146	-0.0002	0.2535	0.3391	0.6609	0.0001	0.0001	0.0001	0.7389	0.4627	Yes	-985.96
Plug-in	-0.0001	0.0219	0.0006	0.1418	0.0001	0.0001	0.0001	0.1021	0.2768	0.7229	0.4883	0.7502	Yes	-726.36
Quadrature	0.0917	-0.0001	0.1461	-0.0000	0.1861	0.3089	0.6911	0.0001	0.0001	0.0001	0.7506	0.4853	Yes	-1012.80
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-3006.88

Exogenous spec; pivot = $^GSPC.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0213 + 0.2146x_t \quad (21)$$

$$\mu_2 = -0.0001 - 0.0002x_t \quad (22)$$

$$\sigma_t^2(1) = 0.2535 + 0.3391 y_{t-1}^2 + 0.6609 \sigma_{t-1}^2 \quad (23)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (24)$$

$$P = \begin{pmatrix} 0.7389 & 0.2611 \\ 0.5373 & 0.4627 \end{pmatrix} \quad (25)$$

LogLik = -985.96; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0001 + 0.0006 x_t \quad (26)$$

$$\mu_2 = 0.0219 + 0.1418 x_t \quad (27)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (28)$$

$$\sigma_t^2(2) = 0.1021 + 0.2768 y_{t-1}^2 + 0.7229 \sigma_{t-1}^2 \quad (29)$$

$$P = \begin{pmatrix} 0.4883 & 0.5117 \\ 0.2498 & 0.7502 \end{pmatrix} \quad (30)$$

LogLik = -726.36; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0917 + 0.1461 x_t \quad (31)$$

$$\mu_2 = -0.0001 - 0.0000 x_t \quad (32)$$

$$\sigma_t^2(1) = 0.1861 + 0.3089 y_{t-1}^2 + 0.6911 \sigma_{t-1}^2 \quad (33)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (34)$$

$$P = \begin{pmatrix} 0.7506 & 0.2494 \\ 0.5147 & 0.4853 \end{pmatrix} \quad (35)$$

LogLik = -1012.80; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000 x_t \quad (36)$$

$$\mu_2 = 0.0000 + 0.1000 x_t \quad (37)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (38)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (39)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (40)$$

LogLik = -3006.88; Converged: No.

1.2. Asset: N225**Baseline Specification****Table 3: N225 — Baseline Specification**

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0588	0.0001	–	–	0.5097	0.1881	0.8032	0.0001	0.0001	0.0001	0.7047	0.4934	Yes	-1523.48
Plug-in	-0.0002	0.0666	–	–	0.0001	0.0001	0.0001	0.1062	0.1389	0.8444	0.5296	0.7529	Yes	-1183.10
Quadrature	0.1001	0.0003	–	–	0.6615	0.2244	0.7755	0.0001	0.0001	0.0001	0.6441	0.6348	Yes	-1617.53
Particle Filter	-0.0000	-0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3946.35

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):**Gray's Method**

$$\mu_1 = 0.0588 \quad (41)$$

$$\mu_2 = 0.0001 \quad (42)$$

$$\sigma_t^2(1) = 0.5097 + 0.1881 y_{t-1}^2 + 0.8032 \sigma_{t-1}^2 \quad (43)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (44)$$

$$P = \begin{pmatrix} 0.7047 & 0.2953 \\ 0.5066 & 0.4934 \end{pmatrix} \quad (45)$$

LogLik = -1523.48; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0002 \quad (46)$$

$$\mu_2 = 0.0666 \quad (47)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (48)$$

$$\sigma_t^2(2) = 0.1062 + 0.1389 y_{t-1}^2 + 0.8444 \sigma_{t-1}^2 \quad (49)$$

$$P = \begin{pmatrix} 0.5296 & 0.4704 \\ 0.2471 & 0.7529 \end{pmatrix} \quad (50)$$

LogLik = -1183.10; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.1001 \quad (51)$$

$$\mu_2 = 0.0003 \quad (52)$$

$$\sigma_t^2(1) = 0.6615 + 0.2244 y_{t-1}^2 + 0.7755 \sigma_{t-1}^2 \quad (53)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (54)$$

$$P = \begin{pmatrix} 0.6441 & 0.3559 \\ 0.3652 & 0.6348 \end{pmatrix} \quad (55)$$

LogLik = -1617.53; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (56)$$

$$\mu_2 = -0.0000 \quad (57)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (58)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (59)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (60)$$

LogLik = -3946.35; Converged: Yes.

Exogenous Specification**Table 4:** N225 — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0379	-0.0429	0.4057	0.2095	0.7990	0.0975	0.9000	0.0253	0.1172	0.0074	0.2865	0.3852	No	-3164.01
Plug-in	0.0257	-0.0001	0.6279	-0.0002	0.1280	0.1767	0.7711	0.0001	0.0001	0.0001	0.7530	0.5292	Yes	-958.26
Quadrature	0.0163	-0.0003	0.6281	0.0001	0.2268	0.1732	0.8266	0.0001	0.0001	0.0001	0.7203	0.4536	Yes	-1282.22
Particle Filter	-0.0000	-0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3844.16

Exogenous spec; pivot = $^GSPC.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0379 + 0.4057 x_t \quad (61)$$

$$\mu_2 = -0.0429 + 0.2095 x_t \quad (62)$$

$$\sigma_t^2(1) = 0.7990 + 0.0975 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (63)$$

$$\sigma_t^2(2) = 0.0253 + 0.1172 y_{t-1}^2 + 0.0074 \sigma_{t-1}^2 \quad (64)$$

$$P = \begin{pmatrix} 0.2865 & 0.7135 \\ 0.6148 & 0.3852 \end{pmatrix} \quad (65)$$

LogLik = -3164.01; Converged: No.

Plug-in Method

$$\mu_1 = 0.0257 + 0.6279 x_t \quad (66)$$

$$\mu_2 = -0.0001 - 0.0002 x_t \quad (67)$$

$$\sigma_t^2(1) = 0.1280 + 0.1767 y_{t-1}^2 + 0.7711 \sigma_{t-1}^2 \quad (68)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (69)$$

$$P = \begin{pmatrix} 0.7530 & 0.2470 \\ 0.4708 & 0.5292 \end{pmatrix} \quad (70)$$

LogLik = -958.26; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0163 + 0.6281 x_t \quad (71)$$

$$\mu_2 = -0.0003 + 0.0001 x_t \quad (72)$$

$$\sigma_t^2(1) = 0.2268 + 0.1732 y_{t-1}^2 + 0.8266 \sigma_{t-1}^2 \quad (73)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (74)$$

$$P = \begin{pmatrix} 0.7203 & 0.2797 \\ 0.5464 & 0.4536 \end{pmatrix} \quad (75)$$

LogLik = -1282.22; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 + 0.1000 x_t \quad (76)$$

$$\mu_2 = -0.0000 + 0.1000 x_t \quad (77)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (78)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (79)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (80)$$

LogLik = -3844.16; Converged: Yes.

1.3. Asset: IXIC

Baseline Specification

Table 5: IXIC — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.4857	0.1456	—	—	0.6551	0.0766	0.9000	0.0001	0.0001	0.7559	0.8691	0.9569	Yes	-4070.86
Plug-in	0.0001	0.1090	—	—	0.0001	0.0001	0.0004	0.1851	0.2735	0.7206	0.5118	0.7568	No	-1647.97
Quadrature	0.0232	0.1347	—	—	0.2404	0.1729	0.8240	0.0135	0.0737	0.6214	0.8661	0.8884	No	-4186.05
Particle Filter	0.0000	0.0000	—	—	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	No	-4274.18

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.4857 \quad (81)$$

$$\mu_2 = 0.1456 \quad (82)$$

$$\sigma_t^2(1) = 0.6551 + 0.0766 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (83)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.7559 \sigma_{t-1}^2 \quad (84)$$

$$P = \begin{pmatrix} 0.8691 & 0.1309 \\ 0.0431 & 0.9569 \end{pmatrix} \quad (85)$$

LogLik = -4070.86; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0001 \quad (86)$$

$$\mu_2 = 0.1090 \quad (87)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0004 \sigma_{t-1}^2 \quad (88)$$

$$\sigma_t^2(2) = 0.1851 + 0.2735 y_{t-1}^2 + 0.7206 \sigma_{t-1}^2 \quad (89)$$

$$P = \begin{pmatrix} 0.5118 & 0.4882 \\ 0.2432 & 0.7568 \end{pmatrix} \quad (90)$$

LogLik = -1647.97; Converged: No.

Quadrature Method

$$\mu_1 = 0.0232 \quad (91)$$

$$\mu_2 = 0.1347 \quad (92)$$

$$\sigma_t^2(1) = 0.2404 + 0.1729 y_{t-1}^2 + 0.8240 \sigma_{t-1}^2 \quad (93)$$

$$\sigma_t^2(2) = 0.0135 + 0.0737 y_{t-1}^2 + 0.6214 \sigma_{t-1}^2 \quad (94)$$

$$P = \begin{pmatrix} 0.8661 & 0.1339 \\ 0.1116 & 0.8884 \end{pmatrix} \quad (95)$$

LogLik = -4186.05; Converged: No.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (96)$$

$$\mu_2 = 0.0000 \quad (97)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (98)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (99)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (100)$$

LogLik = -4274.18; Converged: No.

Exogenous Specification

Table 6: IXIC — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0509	0.1285	-0.0714	-0.0721	0.2420	0.1588	0.8411	0.0001	0.0949	0.6968	0.8581	0.8672	No	-4155.59
Plug-in	0.0001	0.0711	-0.0013	-0.1073	0.0001	0.0001	0.0004	0.1963	0.2551	0.7449	0.4497	0.7301	Yes	-1647.60
Quadrature	0.1181	0.0718	-0.2952	-0.0036	0.5152	0.2517	0.7483	0.0871	0.1286	0.3366	0.8120	0.8530	Yes	-4192.55
Particle Filter	0.0000	0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-4279.14

Exogenous spec; pivot = $^GSPC.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0509 - 0.0714 x_t \quad (101)$$

$$\mu_2 = 0.1285 - 0.0721 x_t \quad (102)$$

$$\sigma_t^2(1) = 0.2420 + 0.1588 y_{t-1}^2 + 0.8411 \sigma_{t-1}^2 \quad (103)$$

$$\sigma_t^2(2) = 0.0001 + 0.0949 y_{t-1}^2 + 0.6968 \sigma_{t-1}^2 \quad (104)$$

$$P = \begin{pmatrix} 0.8581 & 0.1419 \\ 0.1328 & 0.8672 \end{pmatrix} \quad (105)$$

LogLik = -4155.59; Converged: No.

Plug-in Method

$$\mu_1 = 0.0001 - 0.0013 x_t \quad (106)$$

$$\mu_2 = 0.0711 - 0.1073 x_t \quad (107)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0004 \sigma_{t-1}^2 \quad (108)$$

$$\sigma_t^2(2) = 0.1963 + 0.2551 y_{t-1}^2 + 0.7449 \sigma_{t-1}^2 \quad (109)$$

$$P = \begin{pmatrix} 0.4497 & 0.5503 \\ 0.2699 & 0.7301 \end{pmatrix} \quad (110)$$

LogLik = -1647.60; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.1181 - 0.2952 x_t \quad (111)$$

$$\mu_2 = 0.0718 - 0.0036 x_t \quad (112)$$

$$\sigma_t^2(1) = 0.5152 + 0.2517 y_{t-1}^2 + 0.7483 \sigma_{t-1}^2 \quad (113)$$

$$\sigma_t^2(2) = 0.0871 + 0.1286 y_{t-1}^2 + 0.3366 \sigma_{t-1}^2 \quad (114)$$

$$P = \begin{pmatrix} 0.8120 & 0.1880 \\ 0.1470 & 0.8530 \end{pmatrix} \quad (115)$$

LogLik = -4192.55; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500 x_t \quad (116)$$

$$\mu_2 = 0.0000 + 0.0500 x_t \quad (117)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (118)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (119)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (120)$$

LogLik = -4279.14; Converged: Yes.

1.4. Asset: EEM

Baseline Specification

Table 7: EEM — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0113	0.0050	–	–	0.7411	0.1723	0.8276	0.0001	0.0001	0.1966	0.6085	0.6493	No	-3672.50
Plug-in	0.0353	-0.0000	–	–	0.0657	0.1191	0.8789	0.0001	0.0001	0.0001	0.7691	0.5141	Yes	-1451.69
Quadrature	0.0501	0.0001	–	–	0.3670	0.1404	0.8589	0.0001	0.0001	0.0001	0.7578	0.4318	Yes	-1804.79
Particle Filter	-0.0000	-0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3993.87

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.0113 \quad (121)$$

$$\mu_2 = 0.0050 \quad (122)$$

$$\sigma_t^2(1) = 0.7411 + 0.1723 y_{t-1}^2 + 0.8276 \sigma_{t-1}^2 \quad (123)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.1966 \sigma_{t-1}^2 \quad (124)$$

$$P = \begin{pmatrix} 0.6085 & 0.3915 \\ 0.3507 & 0.6493 \end{pmatrix} \quad (125)$$

LogLik = -3672.50 ; Converged: No.

Plug-in Method

$$\mu_1 = 0.0353 \quad (126)$$

$$\mu_2 = -0.0000 \quad (127)$$

$$\sigma_t^2(1) = 0.0657 + 0.1191 y_{t-1}^2 + 0.8789 \sigma_{t-1}^2 \quad (128)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (129)$$

$$P = \begin{pmatrix} 0.7691 & 0.2309 \\ 0.4859 & 0.5141 \end{pmatrix} \quad (130)$$

LogLik = -1451.69 ; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0501 \quad (131)$$

$$\mu_2 = 0.0001 \quad (132)$$

$$\sigma_t^2(1) = 0.3670 + 0.1404 y_{t-1}^2 + 0.8589 \sigma_{t-1}^2 \quad (133)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (134)$$

$$P = \begin{pmatrix} 0.7578 & 0.2422 \\ 0.5682 & 0.4318 \end{pmatrix} \quad (135)$$

LogLik = -1804.79; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (136)$$

$$\mu_2 = -0.0000 \quad (137)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (138)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (139)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (140)$$

LogLik = -3993.87; Converged: Yes.

Exogenous Specification

Table 8: EEM — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0112	0.0002	-0.0946	0.0003	0.4381	0.1825	0.8134	0.0001	0.0001	0.0001	0.7508	0.4697	Yes	-1777.71
Plug-in	-0.0000	0.0350	-0.0001	-0.0749	0.0001	0.0001	0.0001	0.0643	0.1195	0.8796	0.5192	0.7688	Yes	-1448.70
Quadrature	-0.1344	0.0001	-0.2235	-0.0039	0.8549	0.6400	0.3581	0.0001	0.1089	0.0001	0.5021	0.4844	Yes	-2441.71
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-4011.86

Exogenous spec; pivot = $^G SPC.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0112 - 0.0946 x_t \quad (141)$$

$$\mu_2 = 0.0002 + 0.0003 x_t \quad (142)$$

$$\sigma_t^2(1) = 0.4381 + 0.1825y_{t-1}^2 + 0.8134\sigma_{t-1}^2 \quad (143)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (144)$$

$$P = \begin{pmatrix} 0.7508 & 0.2492 \\ 0.5303 & 0.4697 \end{pmatrix} \quad (145)$$

LogLik = -1777.71; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0000 - 0.0001x_t \quad (146)$$

$$\mu_2 = 0.0350 - 0.0749x_t \quad (147)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (148)$$

$$\sigma_t^2(2) = 0.0643 + 0.1195y_{t-1}^2 + 0.8796\sigma_{t-1}^2 \quad (149)$$

$$P = \begin{pmatrix} 0.5192 & 0.4808 \\ 0.2312 & 0.7688 \end{pmatrix} \quad (150)$$

LogLik = -1448.70; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.1344 - 0.2235x_t \quad (151)$$

$$\mu_2 = 0.0001 - 0.0039x_t \quad (152)$$

$$\sigma_t^2(1) = 0.8549 + 0.6400y_{t-1}^2 + 0.3581\sigma_{t-1}^2 \quad (153)$$

$$\sigma_t^2(2) = 0.0001 + 0.1089y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (154)$$

$$P = \begin{pmatrix} 0.5021 & 0.4979 \\ 0.5156 & 0.4844 \end{pmatrix} \quad (155)$$

LogLik = -2441.71; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000x_t \quad (156)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (157)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (158)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (159)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (160)$$

LogLik = -4011.86; Converged: Yes.

1.5. Asset: VTV

Baseline Specification

Table 9: VTV — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0731	-0.0000	—	—	0.2929	0.3022	0.6977	0.0001	0.0001	0.0001	0.7304	0.4624	Yes	-1125.12
Plug-in	0.0724	-0.0000	—	—	0.1410	0.3708	0.6292	0.0001	0.0001	0.0004	0.7878	0.5170	Yes	-835.80
Quadrature	0.0630	-0.0003	—	—	0.2014	0.3257	0.6729	0.0001	0.0001	0.0001	0.7729	0.4669	Yes	-1161.13
Particle Filter	0.0000	0.0000	—	—	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-3185.81

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0731 \quad (161)$$

$$\mu_2 = -0.0000 \quad (162)$$

$$\sigma_t^2(1) = 0.2929 + 0.3022 y_{t-1}^2 + 0.6977 \sigma_{t-1}^2 \quad (163)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (164)$$

$$P = \begin{pmatrix} 0.7304 & 0.2696 \\ 0.5376 & 0.4624 \end{pmatrix} \quad (165)$$

LogLik = -1125.12; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0724 \quad (166)$$

$$\mu_2 = -0.0000 \quad (167)$$

$$\sigma_t^2(1) = 0.1410 + 0.3708 y_{t-1}^2 + 0.6292 \sigma_{t-1}^2 \quad (168)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0004 \sigma_{t-1}^2 \quad (169)$$

$$P = \begin{pmatrix} 0.7878 & 0.2122 \\ 0.4830 & 0.5170 \end{pmatrix} \quad (170)$$

LogLik = -835.80; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0630 \quad (171)$$

$$\mu_2 = -0.0003 \quad (172)$$

$$\sigma_t^2(1) = 0.2014 + 0.3257 y_{t-1}^2 + 0.6729 \sigma_{t-1}^2 \quad (173)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (174)$$

$$P = \begin{pmatrix} 0.7729 & 0.2271 \\ 0.5331 & 0.4669 \end{pmatrix} \quad (175)$$

LogLik = -1161.13; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (176)$$

$$\mu_2 = 0.0000 \quad (177)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (178)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (179)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (180)$$

LogLik = -3185.81; Converged: No.

Exogenous Specification**Table 10:** VTV — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0869	-0.0228	-0.0107	0.0176	0.5310	0.2426	0.7556	0.0003	0.0064	0.0007	0.7299	0.6145	Yes	-2103.47
Plug-in	-0.0001	0.0459	-0.0005	-0.0119	0.0001	0.0001	0.0005	0.1423	0.3254	0.6741	0.5232	0.7447	Yes	-829.83
Quadrature	0.0734	-0.0000	-0.0054	-0.0003	0.2588	0.3504	0.6496	0.0001	0.0001	0.0001	0.7387	0.4614	Yes	-1164.80
Particle Filter	-0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3197.38

Exogenous spec; pivot = $^G SPC.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0869 - 0.0107 x_t \quad (181)$$

$$\mu_2 = -0.0228 + 0.0176 x_t \quad (182)$$

$$\sigma_t^2(1) = 0.5310 + 0.2426y_{t-1}^2 + 0.7556\sigma_{t-1}^2 \quad (183)$$

$$\sigma_t^2(2) = 0.0003 + 0.0064y_{t-1}^2 + 0.0007\sigma_{t-1}^2 \quad (184)$$

$$P = \begin{pmatrix} 0.7299 & 0.2701 \\ 0.3855 & 0.6145 \end{pmatrix} \quad (185)$$

LogLik = -2103.47; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0001 - 0.0005x_t \quad (186)$$

$$\mu_2 = 0.0459 - 0.0119x_t \quad (187)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0005\sigma_{t-1}^2 \quad (188)$$

$$\sigma_t^2(2) = 0.1423 + 0.3254y_{t-1}^2 + 0.6741\sigma_{t-1}^2 \quad (189)$$

$$P = \begin{pmatrix} 0.5232 & 0.4768 \\ 0.2553 & 0.7447 \end{pmatrix} \quad (190)$$

LogLik = -829.83; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0734 - 0.0054x_t \quad (191)$$

$$\mu_2 = -0.0000 - 0.0003x_t \quad (192)$$

$$\sigma_t^2(1) = 0.2588 + 0.3504y_{t-1}^2 + 0.6496\sigma_{t-1}^2 \quad (193)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (194)$$

$$P = \begin{pmatrix} 0.7387 & 0.2613 \\ 0.5386 & 0.4614 \end{pmatrix} \quad (195)$$

LogLik = -1164.80; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 + 0.1000x_t \quad (196)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (197)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (198)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (199)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (200)$$

LogLik = -3197.38; Converged: Yes.

1.6. Asset: GSPC

Baseline Specification

Table 11: GSPC — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.1151	-0.3419	–	–	0.0001	0.0787	0.7068	0.4046	0.2466	0.7534	0.9516	0.8362	Yes	-3333.75
Plug-in	-0.0001	0.0739	–	–	0.0001	0.0001	0.0001	0.1646	0.3090	0.6910	0.5407	0.7831	No	-1054.82
Quadrature	0.0765	-0.0002	–	–	0.3227	0.3107	0.6891	0.0001	0.0001	0.0001	0.7184	0.4804	No	-1414.35
Particle Filter	-0.0000	-0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3516.35

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.1151 \quad (201)$$

$$\mu_2 = -0.3419 \quad (202)$$

$$\sigma_t^2(1) = 0.0001 + 0.0787 y_{t-1}^2 + 0.7068 \sigma_{t-1}^2 \quad (203)$$

$$\sigma_t^2(2) = 0.4046 + 0.2466 y_{t-1}^2 + 0.7534 \sigma_{t-1}^2 \quad (204)$$

$$P = \begin{pmatrix} 0.9516 & 0.0484 \\ 0.1638 & 0.8362 \end{pmatrix} \quad (205)$$

LogLik = -3333.75; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0001 \quad (206)$$

$$\mu_2 = 0.0739 \quad (207)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (208)$$

$$\sigma_t^2(2) = 0.1646 + 0.3090 y_{t-1}^2 + 0.6910 \sigma_{t-1}^2 \quad (209)$$

$$P = \begin{pmatrix} 0.5407 & 0.4593 \\ 0.2169 & 0.7831 \end{pmatrix} \quad (210)$$

LogLik = -1054.82; Converged: No.

Quadrature Method

$$\mu_1 = 0.0765 \quad (211)$$

$$\mu_2 = -0.0002 \quad (212)$$

$$\sigma_t^2(1) = 0.3227 + 0.3107 y_{t-1}^2 + 0.6891 \sigma_{t-1}^2 \quad (213)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (214)$$

$$P = \begin{pmatrix} 0.7184 & 0.2816 \\ 0.5196 & 0.4804 \end{pmatrix} \quad (215)$$

LogLik = -1414.35; Converged: No.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (216)$$

$$\mu_2 = -0.0000 \quad (217)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (218)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (219)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (220)$$

LogLik = -3516.35; Converged: Yes.

2. Bond Market

Pivot (exogenous conditioning variable): TNX.

2.1. Asset: IEI

Baseline Specification

Table 12: IEI — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0019	0.0006	–	–	0.0256	0.2409	0.7561	0.0001	0.0001	0.0001	0.7280	0.5054	Yes	1896.91
Plug-in	0.0129	-0.0000	–	–	0.0017	0.1031	0.8969	0.0001	0.0001	0.0001	0.7409	0.5922	Yes	2031.30
Quadrature	0.0006	0.0001	–	–	0.0001	0.2845	0.5719	0.0001	0.0001	0.0001	0.7160	0.4853	Yes	1892.71
Particle Filter	-0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-1455.01

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0019 \quad (221)$$

$$\mu_2 = 0.0006 \quad (222)$$

$$\sigma_t^2(1) = 0.0256 + 0.2409 y_{t-1}^2 + 0.7561 \sigma_{t-1}^2 \quad (223)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (224)$$

$$P = \begin{pmatrix} 0.7280 & 0.2720 \\ 0.4946 & 0.5054 \end{pmatrix} \quad (225)$$

LogLik = 1896.91; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0129 \quad (226)$$

$$\mu_2 = -0.0000 \quad (227)$$

$$\sigma_t^2(1) = 0.0017 + 0.1031 y_{t-1}^2 + 0.8969 \sigma_{t-1}^2 \quad (228)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (229)$$

$$P = \begin{pmatrix} 0.7409 & 0.2591 \\ 0.4078 & 0.5922 \end{pmatrix} \quad (230)$$

LogLik = 2031.30; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0006 \quad (231)$$

$$\mu_2 = 0.0001 \quad (232)$$

$$\sigma_t^2(1) = 0.0001 + 0.2845 y_{t-1}^2 + 0.5719 \sigma_{t-1}^2 \quad (233)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (234)$$

$$P = \begin{pmatrix} 0.7160 & 0.2840 \\ 0.5147 & 0.4853 \end{pmatrix} \quad (235)$$

LogLik = 1892.71; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (236)$$

$$\mu_2 = 0.0000 \quad (237)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (238)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (239)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (240)$$

LogLik = -1455.01; Converged: Yes.

Exogenous Specification

Table 13: IEI — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0011	-0.0000	0.0026	-0.0000	0.0270	0.3322	0.6635	0.0001	0.0001	0.0001	0.7347	0.4642	Yes	1891.71
Plug-in	0.0002	-0.0085	0.0002	-0.0026	0.0001	0.0001	0.0001	0.0086	0.2878	0.7122	0.5018	0.7526	No	1994.27
Quadrature	0.0005	0.0001	0.0022	0.0001	0.0001	0.2840	0.5712	0.0001	0.0001	0.0001	0.7167	0.4850	Yes	1893.21
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-1693.27

Exogenous spec; pivot = $^T NX.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0011 + 0.0026 x_t \quad (241)$$

$$\mu_2 = -0.0000 - 0.0000 x_t \quad (242)$$

$$\sigma_t^2(1) = 0.0270 + 0.3322y_{t-1}^2 + 0.6635\sigma_{t-1}^2 \quad (243)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (244)$$

$$P = \begin{pmatrix} 0.7347 & 0.2653 \\ 0.5358 & 0.4642 \end{pmatrix} \quad (245)$$

LogLik = 1891.71; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0002 + 0.0002x_t \quad (246)$$

$$\mu_2 = -0.0085 - 0.0026x_t \quad (247)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (248)$$

$$\sigma_t^2(2) = 0.0086 + 0.2878y_{t-1}^2 + 0.7122\sigma_{t-1}^2 \quad (249)$$

$$P = \begin{pmatrix} 0.5018 & 0.4982 \\ 0.2474 & 0.7526 \end{pmatrix} \quad (250)$$

LogLik = 1994.27; Converged: No.

Quadrature Method

$$\mu_1 = 0.0005 + 0.0022x_t \quad (251)$$

$$\mu_2 = 0.0001 + 0.0001x_t \quad (252)$$

$$\sigma_t^2(1) = 0.0001 + 0.2840y_{t-1}^2 + 0.5712\sigma_{t-1}^2 \quad (253)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (254)$$

$$P = \begin{pmatrix} 0.7167 & 0.2833 \\ 0.5150 & 0.4850 \end{pmatrix} \quad (255)$$

LogLik = 1893.21; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000x_t \quad (256)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (257)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (258)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (259)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (260)$$

LogLik = -1693.27; Converged: No.

2.2. Asset: TLT

Baseline Specification

Table 14: TLT — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0204	-0.0001	–	–	0.4310	0.1625	0.5545	0.0001	0.0001	0.0001	0.7467	0.5053	Yes	-1173.63
Plug-in	-0.0082	-0.0001	–	–	0.0678	0.1965	0.8035	0.0001	0.0001	0.0001	0.7908	0.5119	Yes	-934.93
Quadrature	-0.0001	-0.0503	–	–	0.0001	0.0001	0.0001	0.1621	0.1985	0.7949	0.4683	0.7582	Yes	-1200.19
Particle Filter	-0.0000	-0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3249.52

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.0204 \quad (261)$$

$$\mu_2 = -0.0001 \quad (262)$$

$$\sigma_t^2(1) = 0.4310 + 0.1625 y_{t-1}^2 + 0.5545 \sigma_{t-1}^2 \quad (263)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (264)$$

$$P = \begin{pmatrix} 0.7467 & 0.2533 \\ 0.4947 & 0.5053 \end{pmatrix} \quad (265)$$

LogLik = -1173.63 ; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0082 \quad (266)$$

$$\mu_2 = -0.0001 \quad (267)$$

$$\sigma_t^2(1) = 0.0678 + 0.1965 y_{t-1}^2 + 0.8035 \sigma_{t-1}^2 \quad (268)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (269)$$

$$P = \begin{pmatrix} 0.7908 & 0.2092 \\ 0.4881 & 0.5119 \end{pmatrix} \quad (270)$$

LogLik = -934.93 ; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0001 \quad (271)$$

$$\mu_2 = -0.0503 \quad (272)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (273)$$

$$\sigma_t^2(2) = 0.1621 + 0.1985 y_{t-1}^2 + 0.7949 \sigma_{t-1}^2 \quad (274)$$

$$P = \begin{pmatrix} 0.4683 & 0.5317 \\ 0.2418 & 0.7582 \end{pmatrix} \quad (275)$$

LogLik = -1200.19; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (276)$$

$$\mu_2 = -0.0000 \quad (277)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (278)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (279)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (280)$$

LogLik = -3249.52; Converged: Yes.

Exogenous Specification**Table 15:** TLT — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0225	-0.0001	-0.0024	-0.0001	0.2859	0.1930	0.8005	0.0001	0.0001	0.0001	0.7471	0.4607	Yes	-1170.40
Plug-in	0.0405	-0.0005	0.0136	0.0001	0.0405	0.1273	0.8727	0.0001	0.0001	0.0001	0.7469	0.5184	Yes	-926.61
Quadrature	-0.0285	-0.0003	0.0059	0.0000	0.1776	0.1736	0.8243	0.0001	0.0001	0.0001	0.7407	0.4673	Yes	-1193.51
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3339.23

Exogenous spec; pivot = $^T NX.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0225 - 0.0024 x_t \quad (281)$$

$$\mu_2 = -0.0001 - 0.0001 x_t \quad (282)$$

$$\sigma_t^2(1) = 0.2859 + 0.1930y_{t-1}^2 + 0.8005 \sigma_{t-1}^2 \quad (283)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (284)$$

$$P = \begin{pmatrix} 0.7471 & 0.2529 \\ 0.5393 & 0.4607 \end{pmatrix} \quad (285)$$

LogLik = -1170.40; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0405 + 0.0136x_t \quad (286)$$

$$\mu_2 = -0.0005 + 0.0001x_t \quad (287)$$

$$\sigma_t^2(1) = 0.0405 + 0.1273y_{t-1}^2 + 0.8727 \sigma_{t-1}^2 \quad (288)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (289)$$

$$P = \begin{pmatrix} 0.7469 & 0.2531 \\ 0.4816 & 0.5184 \end{pmatrix} \quad (290)$$

LogLik = -926.61; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0285 + 0.0059x_t \quad (291)$$

$$\mu_2 = -0.0003 + 0.0000x_t \quad (292)$$

$$\sigma_t^2(1) = 0.1776 + 0.1736y_{t-1}^2 + 0.8243 \sigma_{t-1}^2 \quad (293)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (294)$$

$$P = \begin{pmatrix} 0.7407 & 0.2593 \\ 0.5327 & 0.4673 \end{pmatrix} \quad (295)$$

LogLik = -1193.51; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000x_t \quad (296)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (297)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (298)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (299)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (300)$$

LogLik = -3339.23; Converged: Yes.

2.3. Asset: BNDX

Baseline Specification

Table 16: BNDX — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0068	0.0000	–	–	0.0262	0.3886	0.6108	0.0001	0.0001	0.0001	0.7316	0.4414	Yes	1854.41
Plug-in	0.0005	0.0082	–	–	0.0001	0.0001	0.0020	0.0076	0.3010	0.6990	0.5175	0.7284	Yes	1992.60
Quadrature	0.0071	0.0004	–	–	0.0001	0.3053	0.5660	0.0001	0.0001	0.0001	0.7077	0.4657	Yes	1864.15
Particle Filter	0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-1463.24

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0068 \quad (301)$$

$$\mu_2 = 0.0000 \quad (302)$$

$$\sigma_t^2(1) = 0.0262 + 0.3886 y_{t-1}^2 + 0.6108 \sigma_{t-1}^2 \quad (303)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (304)$$

$$P = \begin{pmatrix} 0.7316 & 0.2684 \\ 0.5586 & 0.4414 \end{pmatrix} \quad (305)$$

LogLik = 1854.41; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0005 \quad (306)$$

$$\mu_2 = 0.0082 \quad (307)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0020 \sigma_{t-1}^2 \quad (308)$$

$$\sigma_t^2(2) = 0.0076 + 0.3010 y_{t-1}^2 + 0.6990 \sigma_{t-1}^2 \quad (309)$$

$$P = \begin{pmatrix} 0.5175 & 0.4825 \\ 0.2716 & 0.7284 \end{pmatrix} \quad (310)$$

LogLik = 1992.60; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0071 \quad (311)$$

$$\mu_2 = 0.0004 \quad (312)$$

$$\sigma_t^2(1) = 0.0001 + 0.3053 y_{t-1}^2 + 0.5660 \sigma_{t-1}^2 \quad (313)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (314)$$

$$P = \begin{pmatrix} 0.7077 & 0.2923 \\ 0.5343 & 0.4657 \end{pmatrix} \quad (315)$$

LogLik = 1864.15; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (316)$$

$$\mu_2 = 0.0000 \quad (317)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (318)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (319)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (320)$$

LogLik = -1463.24; Converged: No.

Exogenous Specification

Table 17: BNDX — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0072	0.0005	-0.0099	0.0001	0.0249	0.2959	0.7019	0.0001	0.0001	0.0001	0.7255	0.4732	Yes	1874.11
Plug-in	0.0004	0.0098	0.0001	-0.0050	0.0001	0.0001	0.0012	0.0103	0.3731	0.6265	0.4482	0.7258	Yes	1965.75
Quadrature	0.0082	0.0004	-0.0099	0.0001	0.0001	0.2883	0.5664	0.0001	0.0001	0.0001	0.7055	0.4650	Yes	1871.45
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-1707.50

Exogenous spec; pivot = $^T NX.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0072 - 0.0099 x_t \quad (321)$$

$$\mu_2 = 0.0005 + 0.0001 x_t \quad (322)$$

$$\sigma_t^2(1) = 0.0249 + 0.2959y_{t-1}^2 + 0.7019\sigma_{t-1}^2 \quad (323)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (324)$$

$$P = \begin{pmatrix} 0.7255 & 0.2745 \\ 0.5268 & 0.4732 \end{pmatrix} \quad (325)$$

LogLik = 1874.11; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0004 + 0.0001x_t \quad (326)$$

$$\mu_2 = 0.0098 - 0.0050x_t \quad (327)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0012\sigma_{t-1}^2 \quad (328)$$

$$\sigma_t^2(2) = 0.0103 + 0.3731y_{t-1}^2 + 0.6265\sigma_{t-1}^2 \quad (329)$$

$$P = \begin{pmatrix} 0.4482 & 0.5518 \\ 0.2742 & 0.7258 \end{pmatrix} \quad (330)$$

LogLik = 1965.75; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0082 - 0.0099x_t \quad (331)$$

$$\mu_2 = 0.0004 + 0.0001x_t \quad (332)$$

$$\sigma_t^2(1) = 0.0001 + 0.2883y_{t-1}^2 + 0.5664\sigma_{t-1}^2 \quad (333)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (334)$$

$$P = \begin{pmatrix} 0.7055 & 0.2945 \\ 0.5350 & 0.4650 \end{pmatrix} \quad (335)$$

LogLik = 1871.45; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000x_t \quad (336)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (337)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (338)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (339)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (340)$$

LogLik = -1707.50; Converged: No.

2.4. Asset: LQD

Baseline Specification

Table 18: LQD — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0000	0.0001	–	–	0.0001	0.0001	0.0001	0.0791	0.3147	0.6852	0.3967	0.7547	Yes	310.69
Plug-in	-0.0000	0.0325	–	–	0.0001	0.0001	0.0001	0.0295	0.2940	0.7060	0.5094	0.8025	No	482.71
Quadrature	0.0243	-0.0003	–	–	0.0180	0.2731	0.7269	0.0001	0.0001	0.0001	0.7045	0.4639	Yes	293.74
Particle Filter	0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-1996.75

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0000 \quad (341)$$

$$\mu_2 = 0.0001 \quad (342)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (343)$$

$$\sigma_t^2(2) = 0.0791 + 0.3147 y_{t-1}^2 + 0.6852 \sigma_{t-1}^2 \quad (344)$$

$$P = \begin{pmatrix} 0.3967 & 0.6033 \\ 0.2453 & 0.7547 \end{pmatrix} \quad (345)$$

LogLik = 310.69; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0000 \quad (346)$$

$$\mu_2 = 0.0325 \quad (347)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (348)$$

$$\sigma_t^2(2) = 0.0295 + 0.2940 y_{t-1}^2 + 0.7060 \sigma_{t-1}^2 \quad (349)$$

$$P = \begin{pmatrix} 0.5094 & 0.4906 \\ 0.1975 & 0.8025 \end{pmatrix} \quad (350)$$

LogLik = 482.71; Converged: No.

Quadrature Method

$$\mu_1 = 0.0243 \quad (351)$$

$$\mu_2 = -0.0003 \quad (352)$$

$$\sigma_t^2(1) = 0.0180 + 0.2731 y_{t-1}^2 + 0.7269 \sigma_{t-1}^2 \quad (353)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (354)$$

$$P = \begin{pmatrix} 0.7045 & 0.2955 \\ 0.5361 & 0.4639 \end{pmatrix} \quad (355)$$

LogLik = 293.74; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (356)$$

$$\mu_2 = 0.0000 \quad (357)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (358)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (359)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (360)$$

LogLik = -1996.75; Converged: No.

Exogenous Specification

Table 19: LQD — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0044	-0.0000	0.0111	-0.0002	0.0895	0.3175	0.6825	0.0001	0.0001	0.0001	0.7551	0.4700	Yes	316.53
Plug-in	0.0153	0.0001	0.0110	-0.0002	0.0286	0.3169	0.6831	0.0001	0.0001	0.0001	0.8154	0.5081	Yes	467.55
Quadrature	0.0475	0.0000	0.0230	-0.0001	0.0231	0.3555	0.6445	0.0001	0.0001	0.0001	0.7441	0.3965	Yes	275.63
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-2165.86

Exogenous spec; pivot = ${}^T NX.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0044 + 0.0111 x_t \quad (361)$$

$$\mu_2 = -0.0000 - 0.0002 x_t \quad (362)$$

$$\sigma_t^2(1) = 0.0895 + 0.3175y_{t-1}^2 + 0.6825\sigma_{t-1}^2 \quad (363)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (364)$$

$$P = \begin{pmatrix} 0.7551 & 0.2449 \\ 0.5300 & 0.4700 \end{pmatrix} \quad (365)$$

LogLik = 316.53; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0153 + 0.0110x_t \quad (366)$$

$$\mu_2 = 0.0001 - 0.0002x_t \quad (367)$$

$$\sigma_t^2(1) = 0.0286 + 0.3169y_{t-1}^2 + 0.6831\sigma_{t-1}^2 \quad (368)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (369)$$

$$P = \begin{pmatrix} 0.8154 & 0.1846 \\ 0.4919 & 0.5081 \end{pmatrix} \quad (370)$$

LogLik = 467.55; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0475 + 0.0230x_t \quad (371)$$

$$\mu_2 = 0.0000 - 0.0001x_t \quad (372)$$

$$\sigma_t^2(1) = 0.0231 + 0.3555y_{t-1}^2 + 0.6445\sigma_{t-1}^2 \quad (373)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (374)$$

$$P = \begin{pmatrix} 0.7441 & 0.2559 \\ 0.6035 & 0.3965 \end{pmatrix} \quad (375)$$

LogLik = 275.63; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000x_t \quad (376)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (377)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (378)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (379)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (380)$$

LogLik = -2165.86; Converged: No.

2.5. Asset: HYG

Baseline Specification

Table 20: HYG — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0246	-0.0015	–	–	0.0673	0.4066	0.5934	0.0001	0.0001	0.0001	0.8010	0.4364	No	597.67
Plug-in	0.0015	0.0408	–	–	0.0001	0.0001	0.2165	0.0300	0.3273	0.6727	0.6582	0.7240	Yes	697.38
Quadrature	-0.0001	0.0526	–	–	0.0001	0.0001	0.0001	0.0272	0.5074	0.4926	0.4019	0.7372	Yes	579.36
Particle Filter	0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-1889.53

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0246 \quad (381)$$

$$\mu_2 = -0.0015 \quad (382)$$

$$\sigma_t^2(1) = 0.0673 + 0.4066 y_{t-1}^2 + 0.5934 \sigma_{t-1}^2 \quad (383)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (384)$$

$$P = \begin{pmatrix} 0.8010 & 0.1990 \\ 0.5636 & 0.4364 \end{pmatrix} \quad (385)$$

LogLik = 597.67; Converged: No.

Plug-in Method

$$\mu_1 = 0.0015 \quad (386)$$

$$\mu_2 = 0.0408 \quad (387)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.2165 \sigma_{t-1}^2 \quad (388)$$

$$\sigma_t^2(2) = 0.0300 + 0.3273 y_{t-1}^2 + 0.6727 \sigma_{t-1}^2 \quad (389)$$

$$P = \begin{pmatrix} 0.6582 & 0.3418 \\ 0.2760 & 0.7240 \end{pmatrix} \quad (390)$$

LogLik = 697.38; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0001 \quad (391)$$

$$\mu_2 = 0.0526 \quad (392)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (393)$$

$$\sigma_t^2(2) = 0.0272 + 0.5074 y_{t-1}^2 + 0.4926 \sigma_{t-1}^2 \quad (394)$$

$$P = \begin{pmatrix} 0.4019 & 0.5981 \\ 0.2628 & 0.7372 \end{pmatrix} \quad (395)$$

LogLik = 579.36; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (396)$$

$$\mu_2 = 0.0000 \quad (397)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (398)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (399)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (400)$$

LogLik = -1889.53; Converged: Yes.

Exogenous Specification

Table 21: HYG — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0378	-0.0003	-0.0098	0.0000	0.0831	0.3436	0.6564	0.0001	0.3080	0.0001	0.4548	0.3933	Yes	270.36
Plug-in	0.0201	-0.0001	-0.0124	-0.0000	0.0274	0.4540	0.5460	0.0001	0.0001	0.0001	0.6616	0.4027	No	745.60
Quadrature	0.0320	-0.0001	-0.0110	0.0000	0.0350	0.3466	0.6534	0.0001	0.4030	0.0001	0.4341	0.4188	Yes	260.54
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-2060.93

Exogenous spec; pivot = $^T NX.Conv.$ = *convergencestatus*.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0378 - 0.0098 x_t \quad (401)$$

$$\mu_2 = -0.0003 + 0.0000 x_t \quad (402)$$

$$\sigma_t^2(1) = 0.0831 + 0.3436y_{t-1}^2 + 0.6564\sigma_{t-1}^2 \quad (403)$$

$$\sigma_t^2(2) = 0.0001 + 0.3080y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (404)$$

$$P = \begin{pmatrix} 0.4548 & 0.5452 \\ 0.6067 & 0.3933 \end{pmatrix} \quad (405)$$

LogLik = 270.36; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0201 - 0.0124x_t \quad (406)$$

$$\mu_2 = -0.0001 - 0.0000x_t \quad (407)$$

$$\sigma_t^2(1) = 0.0274 + 0.4540y_{t-1}^2 + 0.5460\sigma_{t-1}^2 \quad (408)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (409)$$

$$P = \begin{pmatrix} 0.6616 & 0.3384 \\ 0.5973 & 0.4027 \end{pmatrix} \quad (410)$$

LogLik = 745.60; Converged: No.

Quadrature Method

$$\mu_1 = 0.0320 - 0.0110x_t \quad (411)$$

$$\mu_2 = -0.0001 + 0.0000x_t \quad (412)$$

$$\sigma_t^2(1) = 0.0350 + 0.3466y_{t-1}^2 + 0.6534\sigma_{t-1}^2 \quad (413)$$

$$\sigma_t^2(2) = 0.0001 + 0.4030y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (414)$$

$$P = \begin{pmatrix} 0.4341 & 0.5659 \\ 0.5812 & 0.4188 \end{pmatrix} \quad (415)$$

LogLik = 260.54; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000x_t \quad (416)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (417)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (418)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (419)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (420)$$

LogLik = -2060.93; Converged: Yes.

2.6. Asset: TNX

Baseline Specification

Table 22: TNX — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.1184	-0.1153	–	–	0.2283	0.1209	0.8788	0.0404	0.0723	0.7736	0.9332	0.8257	No	-5744.47
Plug-in	0.0895	0.1516	–	–	0.1531	0.1622	0.8344	0.2504	0.2472	0.7193	0.9065	0.8745	No	-5752.84
Quadrature	-0.1040	0.0193	–	–	0.3746	0.0622	0.7342	0.0237	0.0999	0.9000	0.7449	0.9900	Yes	-5708.41
Particle Filter	-0.0000	-0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-5821.83

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.1184 \quad (421)$$

$$\mu_2 = -0.1153 \quad (422)$$

$$\sigma_t^2(1) = 0.2283 + 0.1209 y_{t-1}^2 + 0.8788 \sigma_{t-1}^2 \quad (423)$$

$$\sigma_t^2(2) = 0.0404 + 0.0723 y_{t-1}^2 + 0.7736 \sigma_{t-1}^2 \quad (424)$$

$$P = \begin{pmatrix} 0.9332 & 0.0668 \\ 0.1743 & 0.8257 \end{pmatrix} \quad (425)$$

LogLik = -5744.47 ; Converged: No.

Plug-in Method

$$\mu_1 = 0.0895 \quad (426)$$

$$\mu_2 = 0.1516 \quad (427)$$

$$\sigma_t^2(1) = 0.1531 + 0.1622 y_{t-1}^2 + 0.8344 \sigma_{t-1}^2 \quad (428)$$

$$\sigma_t^2(2) = 0.2504 + 0.2472 y_{t-1}^2 + 0.7193 \sigma_{t-1}^2 \quad (429)$$

$$P = \begin{pmatrix} 0.9065 & 0.0935 \\ 0.1255 & 0.8745 \end{pmatrix} \quad (430)$$

LogLik = -5752.84 ; Converged: No.

Quadrature Method

$$\mu_1 = -0.1040 \quad (431)$$

$$\mu_2 = 0.0193 \quad (432)$$

$$\sigma_t^2(1) = 0.3746 + 0.0622 y_{t-1}^2 + 0.7342 \sigma_{t-1}^2 \quad (433)$$

$$\sigma_t^2(2) = 0.0237 + 0.0999 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (434)$$

$$P = \begin{pmatrix} 0.7449 & 0.2551 \\ 0.0100 & 0.9900 \end{pmatrix} \quad (435)$$

LogLik = -5708.41; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (436)$$

$$\mu_2 = -0.0000 \quad (437)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (438)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (439)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (440)$$

LogLik = -5821.83; Converged: Yes.

3. Cryptocurrency Market

Pivot (exogenous conditioning variable): BTC-USD.

3.1. Asset: ETH-USD

Baseline Specification

Table 23: ETH-USD — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.6126	-0.6393	–	–	0.5422	0.0598	0.9000	0.3453	0.1928	0.8057	0.9045	0.7191	Yes	-8106.68
Plug-in	0.3134	0.1208	–	–	0.4091	0.1081	0.8919	0.2186	0.1249	0.7828	0.9506	0.7508	No	-8073.57
Quadrature	0.4393	-0.4404	–	–	0.4501	0.1080	0.8919	0.2665	0.1473	0.7954	0.9554	0.7309	Yes	-8120.74
Particle Filter	0.0000	-0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-9075.70

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.6126 \quad (441)$$

$$\mu_2 = -0.6393 \quad (442)$$

$$\sigma_t^2(1) = 0.5422 + 0.0598 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (443)$$

$$\sigma_t^2(2) = 0.3453 + 0.1928 y_{t-1}^2 + 0.8057 \sigma_{t-1}^2 \quad (444)$$

$$P = \begin{pmatrix} 0.9045 & 0.0955 \\ 0.2809 & 0.7191 \end{pmatrix} \quad (445)$$

LogLik = -8106.68; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.3134 \quad (446)$$

$$\mu_2 = 0.1208 \quad (447)$$

$$\sigma_t^2(1) = 0.4091 + 0.1081 y_{t-1}^2 + 0.8919 \sigma_{t-1}^2 \quad (448)$$

$$\sigma_t^2(2) = 0.2186 + 0.1249 y_{t-1}^2 + 0.7828 \sigma_{t-1}^2 \quad (449)$$

$$P = \begin{pmatrix} 0.9506 & 0.0494 \\ 0.2492 & 0.7508 \end{pmatrix} \quad (450)$$

LogLik = -8073.57; Converged: No.

Quadrature Method

$$\mu_1 = 0.4393 \quad (451)$$

$$\mu_2 = -0.4404 \quad (452)$$

$$\sigma_t^2(1) = 0.4501 + 0.1080 y_{t-1}^2 + 0.8919 \sigma_{t-1}^2 \quad (453)$$

$$\sigma_t^2(2) = 0.2665 + 0.1473 y_{t-1}^2 + 0.7954 \sigma_{t-1}^2 \quad (454)$$

$$P = \begin{pmatrix} 0.9554 & 0.0446 \\ 0.2691 & 0.7309 \end{pmatrix} \quad (455)$$

LogLik = -8120.74; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (456)$$

$$\mu_2 = -0.0000 \quad (457)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (458)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (459)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (460)$$

LogLik = -9075.70; Converged: Yes.

Exogenous Specification**Table 24:** ETH-USD — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.4167	-0.5453	-0.0869	0.5502	0.4877	0.0668	0.8936	0.3127	0.1736	0.8255	0.9791	0.7073	Yes	-8072.60
Plug-in	0.5905	0.5930	-0.6630	0.7643	0.6762	0.2424	0.6657	0.5989	0.3196	0.6704	0.7854	0.8105	No	-8295.24
Quadrature	0.2852	-0.3721	-0.0624	0.4278	0.4565	0.0608	0.9000	0.2452	0.1417	0.8569	0.9653	0.7370	Yes	-8072.25
Particle Filter	0.0000	0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-9071.62

Exogenous spec; pivot = BTC-USD. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.4167 - 0.0869 x_t \quad (461)$$

$$\mu_2 = -0.5453 + 0.5502 x_t \quad (462)$$

$$\sigma_t^2(1) = 0.4877 + 0.0668 y_{t-1}^2 + 0.8936 \sigma_{t-1}^2 \quad (463)$$

$$\sigma_t^2(2) = 0.3127 + 0.1736 y_{t-1}^2 + 0.8255 \sigma_{t-1}^2 \quad (464)$$

$$P = \begin{pmatrix} 0.9791 & 0.0209 \\ 0.2927 & 0.7073 \end{pmatrix} \quad (465)$$

LogLik = -8072.60; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.5905 - 0.6630 x_t \quad (466)$$

$$\mu_2 = 0.5930 + 0.7643 x_t \quad (467)$$

$$\sigma_t^2(1) = 0.6762 + 0.2424 y_{t-1}^2 + 0.6657 \sigma_{t-1}^2 \quad (468)$$

$$\sigma_t^2(2) = 0.5989 + 0.3196 y_{t-1}^2 + 0.6704 \sigma_{t-1}^2 \quad (469)$$

$$P = \begin{pmatrix} 0.7854 & 0.2146 \\ 0.1895 & 0.8105 \end{pmatrix} \quad (470)$$

LogLik = -8295.24; Converged: No.

Quadrature Method

$$\mu_1 = 0.2852 - 0.0624 x_t \quad (471)$$

$$\mu_2 = -0.3721 + 0.4278 x_t \quad (472)$$

$$\sigma_t^2(1) = 0.4565 + 0.0608 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (473)$$

$$\sigma_t^2(2) = 0.2452 + 0.1417 y_{t-1}^2 + 0.8569 \sigma_{t-1}^2 \quad (474)$$

$$P = \begin{pmatrix} 0.9653 & 0.0347 \\ 0.2630 & 0.7370 \end{pmatrix} \quad (475)$$

LogLik = -8072.25; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500 x_t \quad (476)$$

$$\mu_2 = 0.0000 + 0.0500 x_t \quad (477)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (478)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (479)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (480)$$

LogLik = -9071.62; Converged: Yes.

3.2. Asset: LTC-USD

Baseline Specification

Table 25: LTC-USD — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.4181	-0.3490	–	–	0.4426	0.1128	0.8867	0.2586	0.1439	0.7944	0.9563	0.7337	No	-8316.54
Plug-in	-0.0339	0.1039	–	–	0.4205	0.1101	0.8898	0.2297	0.1314	0.7840	0.9508	0.7462	No	-8269.21
Quadrature	0.4244	-0.3186	–	–	0.4449	0.1151	0.8848	0.2611	0.1455	0.7943	0.9563	0.7327	No	-8317.06
Particle Filter	-0.0000	0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	No	-9342.67

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.4181 \quad (481)$$

$$\mu_2 = -0.3490 \quad (482)$$

$$\sigma_t^2(1) = 0.4426 + 0.1128 y_{t-1}^2 + 0.8867 \sigma_{t-1}^2 \quad (483)$$

$$\sigma_t^2(2) = 0.2586 + 0.1439 y_{t-1}^2 + 0.7944 \sigma_{t-1}^2 \quad (484)$$

$$P = \begin{pmatrix} 0.9563 & 0.0437 \\ 0.2663 & 0.7337 \end{pmatrix} \quad (485)$$

LogLik = -8316.54; Converged: No.

Plug-in Method

$$\mu_1 = -0.0339 \quad (486)$$

$$\mu_2 = 0.1039 \quad (487)$$

$$\sigma_t^2(1) = 0.4205 + 0.1101 y_{t-1}^2 + 0.8898 \sigma_{t-1}^2 \quad (488)$$

$$\sigma_t^2(2) = 0.2297 + 0.1314 y_{t-1}^2 + 0.7840 \sigma_{t-1}^2 \quad (489)$$

$$P = \begin{pmatrix} 0.9508 & 0.0492 \\ 0.2538 & 0.7462 \end{pmatrix} \quad (490)$$

LogLik = -8269.21; Converged: No.

Quadrature Method

$$\mu_1 = 0.4244 \quad (491)$$

$$\mu_2 = -0.3186 \quad (492)$$

$$\sigma_t^2(1) = 0.4449 + 0.1151 y_{t-1}^2 + 0.8848 \sigma_{t-1}^2 \quad (493)$$

$$\sigma_t^2(2) = 0.2611 + 0.1455 y_{t-1}^2 + 0.7943 \sigma_{t-1}^2 \quad (494)$$

$$P = \begin{pmatrix} 0.9563 & 0.0437 \\ 0.2673 & 0.7327 \end{pmatrix} \quad (495)$$

LogLik = -8317.06; Converged: No.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (496)$$

$$\mu_2 = 0.0000 \quad (497)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (498)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (499)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (500)$$

LogLik = -9342.67; Converged: No.

Exogenous Specification**Table 26:** LTC-USD — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.2039	-0.1757	-0.1141	0.2751	0.4092	0.0845	0.9000	0.1869	0.1355	0.8606	0.9884	0.7561	Yes	-8292.09
Plug-in	-0.6148	0.6157	-0.5239	0.6990	0.3360	0.2033	0.7488	0.4238	0.3039	0.6946	0.8040	0.8736	No	-8471.67
Quadrature	0.1923	-0.1625	-0.1149	0.2713	0.4130	0.0865	0.9000	0.1897	0.1409	0.8585	0.9807	0.7488	Yes	-8291.69
Particle Filter	0.0000	0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	No	-9342.78

Exogenous spec; pivot = BTC-USD. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.2039 - 0.1141 x_t \quad (501)$$

$$\mu_2 = -0.1757 + 0.2751 x_t \quad (502)$$

$$\sigma_t^2(1) = 0.4092 + 0.0845 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (503)$$

$$\sigma_t^2(2) = 0.1869 + 0.1355 y_{t-1}^2 + 0.8606 \sigma_{t-1}^2 \quad (504)$$

$$P = \begin{pmatrix} 0.9884 & 0.0116 \\ 0.2439 & 0.7561 \end{pmatrix} \quad (505)$$

LogLik = -8292.09; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.6148 - 0.5239 x_t \quad (506)$$

$$\mu_2 = 0.6157 + 0.6990 x_t \quad (507)$$

$$\sigma_t^2(1) = 0.3360 + 0.2033 y_{t-1}^2 + 0.7488 \sigma_{t-1}^2 \quad (508)$$

$$\sigma_t^2(2) = 0.4238 + 0.3039 y_{t-1}^2 + 0.6946 \sigma_{t-1}^2 \quad (509)$$

$$P = \begin{pmatrix} 0.8040 & 0.1960 \\ 0.1264 & 0.8736 \end{pmatrix} \quad (510)$$

LogLik = -8471.67; Converged: No.

Quadrature Method

$$\mu_1 = 0.1923 - 0.1149 x_t \quad (511)$$

$$\mu_2 = -0.1625 + 0.2713 x_t \quad (512)$$

$$\sigma_t^2(1) = 0.4130 + 0.0865 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (513)$$

$$\sigma_t^2(2) = 0.1897 + 0.1409 y_{t-1}^2 + 0.8585 \sigma_{t-1}^2 \quad (514)$$

$$P = \begin{pmatrix} 0.9807 & 0.0193 \\ 0.2512 & 0.7488 \end{pmatrix} \quad (515)$$

LogLik = -8291.69; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500 x_t \quad (516)$$

$$\mu_2 = 0.0000 + 0.0500 x_t \quad (517)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (518)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (519)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (520)$$

LogLik = -9342.78; Converged: No.

3.3. Asset: XRP-USD

Baseline Specification

Table 27: XRP-USD — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.4434	0.6430	–	–	0.3379	0.1705	0.7626	0.4197	0.2863	0.7136	0.8160	0.8847	No	-9143.45
Plug-in	0.4248	-0.4242	–	–	0.4469	0.1223	0.8774	0.2607	0.1424	0.7849	0.9489	0.7341	No	-8822.00
Quadrature	1.4539	-1.4449	–	–	0.8731	0.1863	0.8134	0.6844	0.3883	0.5541	0.7274	0.7523	No	-8514.52
Particle Filter	-0.0000	-0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-10373.34

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.4434 \quad (521)$$

$$\mu_2 = 0.6430 \quad (522)$$

$$\sigma_t^2(1) = 0.3379 + 0.1705 y_{t-1}^2 + 0.7626 \sigma_{t-1}^2 \quad (523)$$

$$\sigma_t^2(2) = 0.4197 + 0.2863 y_{t-1}^2 + 0.7136 \sigma_{t-1}^2 \quad (524)$$

$$P = \begin{pmatrix} 0.8160 & 0.1840 \\ 0.1153 & 0.8847 \end{pmatrix} \quad (525)$$

LogLik = -9143.45; Converged: No.

Plug-in Method

$$\mu_1 = 0.4248 \quad (526)$$

$$\mu_2 = -0.4242 \quad (527)$$

$$\sigma_t^2(1) = 0.4469 + 0.1223 y_{t-1}^2 + 0.8774 \sigma_{t-1}^2 \quad (528)$$

$$\sigma_t^2(2) = 0.2607 + 0.1424 y_{t-1}^2 + 0.7849 \sigma_{t-1}^2 \quad (529)$$

$$P = \begin{pmatrix} 0.9489 & 0.0511 \\ 0.2659 & 0.7341 \end{pmatrix} \quad (530)$$

LogLik = -8822.00; Converged: No.

Quadrature Method

$$\mu_1 = 1.4539 \quad (531)$$

$$\mu_2 = -1.4449 \quad (532)$$

$$\sigma_t^2(1) = 0.8731 + 0.1863 y_{t-1}^2 + 0.8134 \sigma_{t-1}^2 \quad (533)$$

$$\sigma_t^2(2) = 0.6844 + 0.3883 y_{t-1}^2 + 0.5541 \sigma_{t-1}^2 \quad (534)$$

$$P = \begin{pmatrix} 0.7274 & 0.2726 \\ 0.2477 & 0.7523 \end{pmatrix} \quad (535)$$

LogLik = -8514.52; Converged: No.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (536)$$

$$\mu_2 = -0.0000 \quad (537)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (538)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (539)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (540)$$

LogLik = -10373.34; Converged: Yes.

Exogenous Specification

Table 28: XRP-USD — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.4513	-0.4648	-0.3028	0.4799	0.5408	0.0350	0.9000	0.2814	0.1950	0.8012	0.9489	0.7324	No	-8520.39
Plug-in	0.4094	-0.4084	-0.3651	-0.3614	0.4399	0.1168	0.8829	0.2545	0.1348	0.7829	0.9479	0.7367	No	-8686.18
Quadrature	0.5024	-0.5475	-0.2190	0.3982	0.5735	0.0455	0.9000	0.3131	0.2130	0.7869	0.9438	0.7205	Yes	-8507.94
Particle Filter	0.0032	0.0032	0.0532	0.0467	0.3011	0.0505	0.8500	0.0999	0.0799	0.7801	0.9500	0.8001	Yes	-10360.16

Exogenous spec; pivot = BTC-USD. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.4513 - 0.3028 x_t \quad (541)$$

$$\mu_2 = -0.4648 + 0.4799 x_t \quad (542)$$

$$\sigma_t^2(1) = 0.5408 + 0.0350y_{t-1}^2 + 0.9000\sigma_{t-1}^2 \quad (543)$$

$$\sigma_t^2(2) = 0.2814 + 0.1950y_{t-1}^2 + 0.8012\sigma_{t-1}^2 \quad (544)$$

$$P = \begin{pmatrix} 0.9489 & 0.0511 \\ 0.2676 & 0.7324 \end{pmatrix} \quad (545)$$

LogLik = -8520.39; Converged: No.

Plug-in Method

$$\mu_1 = 0.4094 - 0.3651x_t \quad (546)$$

$$\mu_2 = -0.4084 - 0.3614x_t \quad (547)$$

$$\sigma_t^2(1) = 0.4399 + 0.1168y_{t-1}^2 + 0.8829\sigma_{t-1}^2 \quad (548)$$

$$\sigma_t^2(2) = 0.2545 + 0.1348y_{t-1}^2 + 0.7829\sigma_{t-1}^2 \quad (549)$$

$$P = \begin{pmatrix} 0.9479 & 0.0521 \\ 0.2633 & 0.7367 \end{pmatrix} \quad (550)$$

LogLik = -8686.18; Converged: No.

Quadrature Method

$$\mu_1 = 0.5024 - 0.2190x_t \quad (551)$$

$$\mu_2 = -0.5475 + 0.3982x_t \quad (552)$$

$$\sigma_t^2(1) = 0.5735 + 0.0455y_{t-1}^2 + 0.9000\sigma_{t-1}^2 \quad (553)$$

$$\sigma_t^2(2) = 0.3131 + 0.2130y_{t-1}^2 + 0.7869\sigma_{t-1}^2 \quad (554)$$

$$P = \begin{pmatrix} 0.9438 & 0.0562 \\ 0.2795 & 0.7205 \end{pmatrix} \quad (555)$$

LogLik = -8507.94; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0032 + 0.0532x_t \quad (556)$$

$$\mu_2 = 0.0032 + 0.0467x_t \quad (557)$$

$$\sigma_t^2(1) = 0.3011 + 0.0505y_{t-1}^2 + 0.8500\sigma_{t-1}^2 \quad (558)$$

$$\sigma_t^2(2) = 0.0999 + 0.0799y_{t-1}^2 + 0.7801\sigma_{t-1}^2 \quad (559)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.1999 & 0.8001 \end{pmatrix} \quad (560)$$

LogLik = -10360.16; Converged: Yes.

3.4. Asset: DOGE-USD

Baseline Specification

Table 29: DOGE-USD — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.4167	0.0367	–	–	0.4397	0.1063	0.8935	0.2577	0.1609	0.8029	0.9631	0.7319	No	-8752.64
Plug-in	0.2567	-0.2505	–	–	0.3865	0.1072	0.8928	0.1955	0.1089	0.7798	0.9476	0.7612	No	-8613.45
Quadrature	0.7571	0.7576	–	–	0.3856	0.2121	0.7378	0.4723	0.3139	0.6809	0.7704	0.9076	No	-9069.31
Particle Filter	0.0000	0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	No	-10260.15

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.4167 \quad (561)$$

$$\mu_2 = 0.0367 \quad (562)$$

$$\sigma_t^2(1) = 0.4397 + 0.1063 y_{t-1}^2 + 0.8935 \sigma_{t-1}^2 \quad (563)$$

$$\sigma_t^2(2) = 0.2577 + 0.1609 y_{t-1}^2 + 0.8029 \sigma_{t-1}^2 \quad (564)$$

$$P = \begin{pmatrix} 0.9631 & 0.0369 \\ 0.2681 & 0.7319 \end{pmatrix} \quad (565)$$

LogLik = -8752.64 ; Converged: No.

Plug-in Method

$$\mu_1 = 0.2567 \quad (566)$$

$$\mu_2 = -0.2505 \quad (567)$$

$$\sigma_t^2(1) = 0.3865 + 0.1072 y_{t-1}^2 + 0.8928 \sigma_{t-1}^2 \quad (568)$$

$$\sigma_t^2(2) = 0.1955 + 0.1089 y_{t-1}^2 + 0.7798 \sigma_{t-1}^2 \quad (569)$$

$$P = \begin{pmatrix} 0.9476 & 0.0524 \\ 0.2388 & 0.7612 \end{pmatrix} \quad (570)$$

LogLik = -8613.45 ; Converged: No.

Quadrature Method

$$\mu_1 = 0.7571 \quad (571)$$

$$\mu_2 = 0.7576 \quad (572)$$

$$\sigma_t^2(1) = 0.3856 + 0.2121 y_{t-1}^2 + 0.7378 \sigma_{t-1}^2 \quad (573)$$

$$\sigma_t^2(2) = 0.4723 + 0.3139 y_{t-1}^2 + 0.6809 \sigma_{t-1}^2 \quad (574)$$

$$P = \begin{pmatrix} 0.7704 & 0.2296 \\ 0.0924 & 0.9076 \end{pmatrix} \quad (575)$$

LogLik = -9069.31; Converged: No.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (576)$$

$$\mu_2 = 0.0000 \quad (577)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (578)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (579)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (580)$$

LogLik = -10260.15; Converged: No.

Exogenous Specification**Table 30:** DOGE-USD — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.3375	0.1429	-0.2900	0.3867	0.4231	0.1150	0.8845	0.2385	0.1427	0.7962	0.9397	0.7458	No	-8743.14
Plug-in	-0.6670	0.6699	-0.5781	0.7548	0.3549	0.2127	0.7381	0.4377	0.3194	0.6804	0.7972	0.8755	No	-8872.36
Quadrature	0.3266	0.1761	-0.2776	0.3775	0.4121	0.1128	0.8872	0.2246	0.1395	0.7981	0.9400	0.7511	Yes	-8741.70
Particle Filter	0.0000	0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-10266.08

Exogenous spec; pivot = BTC-USD. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.3375 - 0.2900 x_t \quad (581)$$

$$\mu_2 = 0.1429 + 0.3867 x_t \quad (582)$$

$$\sigma_t^2(1) = 0.4231 + 0.1150y_{t-1}^2 + 0.8845\sigma_{t-1}^2 \quad (583)$$

$$\sigma_t^2(2) = 0.2385 + 0.1427y_{t-1}^2 + 0.7962\sigma_{t-1}^2 \quad (584)$$

$$P = \begin{pmatrix} 0.9397 & 0.0603 \\ 0.2542 & 0.7458 \end{pmatrix} \quad (585)$$

LogLik = -8743.14; Converged: No.

Plug-in Method

$$\mu_1 = -0.6670 - 0.5781x_t \quad (586)$$

$$\mu_2 = 0.6699 + 0.7548x_t \quad (587)$$

$$\sigma_t^2(1) = 0.3549 + 0.2127y_{t-1}^2 + 0.7381\sigma_{t-1}^2 \quad (588)$$

$$\sigma_t^2(2) = 0.4377 + 0.3194y_{t-1}^2 + 0.6804\sigma_{t-1}^2 \quad (589)$$

$$P = \begin{pmatrix} 0.7972 & 0.2028 \\ 0.1245 & 0.8755 \end{pmatrix} \quad (590)$$

LogLik = -8872.36; Converged: No.

Quadrature Method

$$\mu_1 = 0.3266 - 0.2776x_t \quad (591)$$

$$\mu_2 = 0.1761 + 0.3775x_t \quad (592)$$

$$\sigma_t^2(1) = 0.4121 + 0.1128y_{t-1}^2 + 0.8872\sigma_{t-1}^2 \quad (593)$$

$$\sigma_t^2(2) = 0.2246 + 0.1395y_{t-1}^2 + 0.7981\sigma_{t-1}^2 \quad (594)$$

$$P = \begin{pmatrix} 0.9400 & 0.0600 \\ 0.2489 & 0.7511 \end{pmatrix} \quad (595)$$

LogLik = -8741.70; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500x_t \quad (596)$$

$$\mu_2 = 0.0000 + 0.0500x_t \quad (597)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500y_{t-1}^2 + 0.8500\sigma_{t-1}^2 \quad (598)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800y_{t-1}^2 + 0.7800\sigma_{t-1}^2 \quad (599)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (600)$$

LogLik = -10266.08; Converged: Yes.

3.5. Asset: NMC-USD

Baseline Specification

Table 31: NMC-USD — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.3516	0.3519	–	–	0.4205	0.1121	0.8861	0.2336	0.1310	0.7922	0.9638	0.7428	No	-9126.19
Plug-in	-0.3969	0.7938	–	–	0.4041	0.2430	0.7560	0.4814	0.2948	0.6342	0.7760	0.8951	No	-9326.54
Quadrature	0.3394	0.3397	–	–	0.4165	0.1107	0.8879	0.2289	0.1276	0.7921	0.9659	0.7444	No	-9121.91
Particle Filter	-0.0000	0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-10525.65

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.3516 \quad (601)$$

$$\mu_2 = 0.3519 \quad (602)$$

$$\sigma_t^2(1) = 0.4205 + 0.1121 y_{t-1}^2 + 0.8861 \sigma_{t-1}^2 \quad (603)$$

$$\sigma_t^2(2) = 0.2336 + 0.1310 y_{t-1}^2 + 0.7922 \sigma_{t-1}^2 \quad (604)$$

$$P = \begin{pmatrix} 0.9638 & 0.0362 \\ 0.2572 & 0.7428 \end{pmatrix} \quad (605)$$

LogLik = -9126.19; Converged: No.

Plug-in Method

$$\mu_1 = -0.3969 \quad (606)$$

$$\mu_2 = 0.7938 \quad (607)$$

$$\sigma_t^2(1) = 0.4041 + 0.2430 y_{t-1}^2 + 0.7560 \sigma_{t-1}^2 \quad (608)$$

$$\sigma_t^2(2) = 0.4814 + 0.2948 y_{t-1}^2 + 0.6342 \sigma_{t-1}^2 \quad (609)$$

$$P = \begin{pmatrix} 0.7760 & 0.2240 \\ 0.1049 & 0.8951 \end{pmatrix} \quad (610)$$

LogLik = -9326.54; Converged: No.

Quadrature Method

$$\mu_1 = 0.3394 \quad (611)$$

$$\mu_2 = 0.3397 \quad (612)$$

$$\sigma_t^2(1) = 0.4165 + 0.1107 y_{t-1}^2 + 0.8879 \sigma_{t-1}^2 \quad (613)$$

$$\sigma_t^2(2) = 0.2289 + 0.1276 y_{t-1}^2 + 0.7921 \sigma_{t-1}^2 \quad (614)$$

$$P = \begin{pmatrix} 0.9659 & 0.0341 \\ 0.2556 & 0.7444 \end{pmatrix} \quad (615)$$

LogLik = -9121.91; Converged: No.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (616)$$

$$\mu_2 = 0.0000 \quad (617)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (618)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (619)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (620)$$

LogLik = -10525.65; Converged: Yes.

Exogenous Specification

Table 32: NMC-USD — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.2912	0.2977	-0.2788	0.4511	0.4738	0.0678	0.9000	0.2611	0.2195	0.7801	0.8732	0.7470	Yes	-9025.39
Plug-in	0.2801	0.3165	0.3636	-0.2702	0.4081	0.1090	0.8907	0.2200	0.1237	0.7828	0.9486	0.7509	No	-9005.87
Quadrature	0.1893	-0.2462	-0.2546	0.9347	0.7896	0.1000	0.9000	0.4749	0.3541	0.3659	0.9724	0.7735	Yes	-8987.10
Particle Filter	0.0000	-0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-10527.21

Exogenous spec; pivot = BTC-USD. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.2912 - 0.2788 x_t \quad (621)$$

$$\mu_2 = 0.2977 + 0.4511 x_t \quad (622)$$

$$\sigma_t^2(1) = 0.4738 + 0.0678 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (623)$$

$$\sigma_t^2(2) = 0.2611 + 0.2195 y_{t-1}^2 + 0.7801 \sigma_{t-1}^2 \quad (624)$$

$$P = \begin{pmatrix} 0.8732 & 0.1268 \\ 0.2530 & 0.7470 \end{pmatrix} \quad (625)$$

LogLik = -9025.39; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.2801 + 0.3636 x_t \quad (626)$$

$$\mu_2 = 0.3165 - 0.2702 x_t \quad (627)$$

$$\sigma_t^2(1) = 0.4081 + 0.1090 y_{t-1}^2 + 0.8907 \sigma_{t-1}^2 \quad (628)$$

$$\sigma_t^2(2) = 0.2200 + 0.1237 y_{t-1}^2 + 0.7828 \sigma_{t-1}^2 \quad (629)$$

$$P = \begin{pmatrix} 0.9486 & 0.0514 \\ 0.2491 & 0.7509 \end{pmatrix} \quad (630)$$

LogLik = -9005.87; Converged: No.

Quadrature Method

$$\mu_1 = 0.1893 - 0.2546 x_t \quad (631)$$

$$\mu_2 = -0.2462 + 0.9347 x_t \quad (632)$$

$$\sigma_t^2(1) = 0.7896 + 0.1000 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (633)$$

$$\sigma_t^2(2) = 0.4749 + 0.3541 y_{t-1}^2 + 0.3659 \sigma_{t-1}^2 \quad (634)$$

$$P = \begin{pmatrix} 0.9724 & 0.0276 \\ 0.2265 & 0.7735 \end{pmatrix} \quad (635)$$

LogLik = -8987.10; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500 x_t \quad (636)$$

$$\mu_2 = -0.0000 + 0.0500 x_t \quad (637)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (638)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (639)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (640)$$

LogLik = -10527.21; Converged: Yes.

3.6. Asset: BTC-USD

Baseline Specification

Table 33: BTC-USD — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.3250	-0.1326	–	–	0.8259	0.0300	0.9000	0.3511	0.3121	0.6871	0.8265	0.7183	Yes	-7297.71
Plug-in	0.2967	0.2183	–	–	0.4060	0.1119	0.8874	0.0915	0.1184	0.7786	0.9415	0.7555	No	-7232.32
Quadrature	0.3018	0.0777	–	–	0.8128	0.0262	0.9000	0.3007	0.3465	0.6485	0.8635	0.7211	No	-7297.61
Particle Filter	0.0000	0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-7853.80

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.3250 \quad (641)$$

$$\mu_2 = -0.1326 \quad (642)$$

$$\sigma_t^2(1) = 0.8259 + 0.0300 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (643)$$

$$\sigma_t^2(2) = 0.3511 + 0.3121 y_{t-1}^2 + 0.6871 \sigma_{t-1}^2 \quad (644)$$

$$P = \begin{pmatrix} 0.8265 & 0.1735 \\ 0.2817 & 0.7183 \end{pmatrix} \quad (645)$$

LogLik = -7297.71; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.2967 \quad (646)$$

$$\mu_2 = 0.2183 \quad (647)$$

$$\sigma_t^2(1) = 0.4060 + 0.1119 y_{t-1}^2 + 0.8874 \sigma_{t-1}^2 \quad (648)$$

$$\sigma_t^2(2) = 0.0915 + 0.1184 y_{t-1}^2 + 0.7786 \sigma_{t-1}^2 \quad (649)$$

$$P = \begin{pmatrix} 0.9415 & 0.0585 \\ 0.2445 & 0.7555 \end{pmatrix} \quad (650)$$

LogLik = -7232.32; Converged: No.

Quadrature Method

$$\mu_1 = 0.3018 \quad (651)$$

$$\mu_2 = 0.0777 \quad (652)$$

$$\sigma_t^2(1) = 0.8128 + 0.0262 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (653)$$

$$\sigma_t^2(2) = 0.3007 + 0.3465 y_{t-1}^2 + 0.6485 \sigma_{t-1}^2 \quad (654)$$

$$P = \begin{pmatrix} 0.8635 & 0.1365 \\ 0.2789 & 0.7211 \end{pmatrix} \quad (655)$$

LogLik = -7297.61; Converged: No.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (656)$$

$$\mu_2 = 0.0000 \quad (657)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (658)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (659)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (660)$$

LogLik = -7853.80; Converged: Yes.

4. Derivatives Market

Pivot (exogenous conditioning variable): BTC=F.

4.1. Asset: ES=F

Baseline Specification

Table 34: ES=F — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.2298	0.1249	–	–	0.3382	0.1647	0.8349	0.0001	0.0788	0.6385	0.8468	0.9184	Yes	-3363.57
Plug-in	0.0393	0.0001	–	–	0.0990	0.2335	0.7656	0.0001	0.0001	0.0001	0.8017	0.4978	Yes	-1095.35
Quadrature	-0.0299	-0.0005	–	–	0.3765	0.2055	0.7937	0.0001	0.0901	0.0001	0.6221	0.4310	No	-2011.93
Particle Filter	-0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3564.02

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.2298 \quad (661)$$

$$\mu_2 = 0.1249 \quad (662)$$

$$\sigma_t^2(1) = 0.3382 + 0.1647 y_{t-1}^2 + 0.8349 \sigma_{t-1}^2 \quad (663)$$

$$\sigma_t^2(2) = 0.0001 + 0.0788 y_{t-1}^2 + 0.6385 \sigma_{t-1}^2 \quad (664)$$

$$P = \begin{pmatrix} 0.8468 & 0.1532 \\ 0.0816 & 0.9184 \end{pmatrix} \quad (665)$$

LogLik = -3363.57; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0393 \quad (666)$$

$$\mu_2 = 0.0001 \quad (667)$$

$$\sigma_t^2(1) = 0.0990 + 0.2335 y_{t-1}^2 + 0.7656 \sigma_{t-1}^2 \quad (668)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (669)$$

$$P = \begin{pmatrix} 0.8017 & 0.1983 \\ 0.5022 & 0.4978 \end{pmatrix} \quad (670)$$

LogLik = -1095.35; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0299 \quad (671)$$

$$\mu_2 = -0.0005 \quad (672)$$

$$\sigma_t^2(1) = 0.3765 + 0.2055 y_{t-1}^2 + 0.7937 \sigma_{t-1}^2 \quad (673)$$

$$\sigma_t^2(2) = 0.0001 + 0.0901 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (674)$$

$$P = \begin{pmatrix} 0.6221 & 0.3779 \\ 0.5690 & 0.4310 \end{pmatrix} \quad (675)$$

LogLik = -2011.93; Converged: No.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (676)$$

$$\mu_2 = 0.0000 \quad (677)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (678)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (679)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (680)$$

LogLik = -3564.02; Converged: Yes.

Exogenous Specification

Table 35: ES=F — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0705	0.1069	-0.0122	0.0036	0.5880	0.2359	0.7640	0.0234	0.1433	0.2650	0.7659	0.8125	Yes	-3375.42
Plug-in	-0.0002	0.0630	-0.0001	-0.0035	0.0001	0.0001	0.0001	0.0966	0.2883	0.7117	0.5117	0.7569	Yes	-1108.91
Quadrature	0.0204	0.0655	-0.0030	0.0038	0.6408	0.2333	0.7666	0.0001	0.0776	0.1548	0.7089	0.7513	No	-3291.80
Particle Filter	0.0000	0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	No	-3789.14

Exogenous spec; pivot = BTC=F. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0705 - 0.0122 x_t \quad (681)$$

$$\mu_2 = 0.1069 + 0.0036 x_t \quad (682)$$

$$\sigma_t^2(1) = 0.5880 + 0.2359y_{t-1}^2 + 0.7640\sigma_{t-1}^2 \quad (683)$$

$$\sigma_t^2(2) = 0.0234 + 0.1433y_{t-1}^2 + 0.2650\sigma_{t-1}^2 \quad (684)$$

$$P = \begin{pmatrix} 0.7659 & 0.2341 \\ 0.1875 & 0.8125 \end{pmatrix} \quad (685)$$

LogLik = -3375.42; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0002 - 0.0001x_t \quad (686)$$

$$\mu_2 = 0.0630 - 0.0035x_t \quad (687)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (688)$$

$$\sigma_t^2(2) = 0.0966 + 0.2883y_{t-1}^2 + 0.7117\sigma_{t-1}^2 \quad (689)$$

$$P = \begin{pmatrix} 0.5117 & 0.4883 \\ 0.2431 & 0.7569 \end{pmatrix} \quad (690)$$

LogLik = -1108.91; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0204 - 0.0030x_t \quad (691)$$

$$\mu_2 = 0.0655 + 0.0038x_t \quad (692)$$

$$\sigma_t^2(1) = 0.6408 + 0.2333y_{t-1}^2 + 0.7666\sigma_{t-1}^2 \quad (693)$$

$$\sigma_t^2(2) = 0.0001 + 0.0776y_{t-1}^2 + 0.1548\sigma_{t-1}^2 \quad (694)$$

$$P = \begin{pmatrix} 0.7089 & 0.2911 \\ 0.2487 & 0.7513 \end{pmatrix} \quad (695)$$

LogLik = -3291.80; Converged: No.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500x_t \quad (696)$$

$$\mu_2 = 0.0000 + 0.0500x_t \quad (697)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500y_{t-1}^2 + 0.8500\sigma_{t-1}^2 \quad (698)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800y_{t-1}^2 + 0.7800\sigma_{t-1}^2 \quad (699)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (700)$$

LogLik = -3789.14; Converged: No.

4.2. Asset: NQ=F**Baseline Specification****Table 36:** NQ=F — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.4956	0.1470	—	—	0.7317	0.0746	0.9000	0.0001	0.0001	0.7779	0.8629	0.9609	Yes	-4168.20
Plug-in	-0.0002	0.1257	—	—	0.0001	0.0001	0.0001	0.2100	0.2449	0.7539	0.5117	0.7429	Yes	-1708.53
Quadrature	-0.2432	0.0004	—	—	0.9614	0.1800	0.8192	0.0352	0.0909	0.1859	0.5977	0.7402	No	-4142.41
Particle Filter	-0.0000	0.0000	—	—	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-4346.95

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):**Gray's Method**

$$\mu_1 = -0.4956 \quad (701)$$

$$\mu_2 = 0.1470 \quad (702)$$

$$\sigma_t^2(1) = 0.7317 + 0.0746 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (703)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.7779 \sigma_{t-1}^2 \quad (704)$$

$$P = \begin{pmatrix} 0.8629 & 0.1371 \\ 0.0391 & 0.9609 \end{pmatrix} \quad (705)$$

LogLik = -4168.20; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0002 \quad (706)$$

$$\mu_2 = 0.1257 \quad (707)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (708)$$

$$\sigma_t^2(2) = 0.2100 + 0.2449 y_{t-1}^2 + 0.7539 \sigma_{t-1}^2 \quad (709)$$

$$P = \begin{pmatrix} 0.5117 & 0.4883 \\ 0.2571 & 0.7429 \end{pmatrix} \quad (710)$$

LogLik = -1708.53; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.2432 \quad (711)$$

$$\mu_2 = 0.0004 \quad (712)$$

$$\sigma_t^2(1) = 0.9614 + 0.1800 y_{t-1}^2 + 0.8192 \sigma_{t-1}^2 \quad (713)$$

$$\sigma_t^2(2) = 0.0352 + 0.0909 y_{t-1}^2 + 0.1859 \sigma_{t-1}^2 \quad (714)$$

$$P = \begin{pmatrix} 0.5977 & 0.4023 \\ 0.2598 & 0.7402 \end{pmatrix} \quad (715)$$

LogLik = -4142.41; Converged: No.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (716)$$

$$\mu_2 = 0.0000 \quad (717)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (718)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (719)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (720)$$

LogLik = -4346.95; Converged: Yes.

Exogenous Specification

Table 37: NQ=F — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0568	0.1104	-0.0123	-0.0051	0.2432	0.1515	0.8484	0.0019	0.0919	0.7138	0.8589	0.8545	No	-4259.78
Plug-in	-0.0001	0.0957	-0.0003	0.0051	0.0001	0.0001	0.0001	0.2453	0.2590	0.7410	0.5016	0.7755	Yes	-1717.33
Quadrature	0.0603	0.1124	-0.0230	0.0022	0.2318	0.1811	0.8172	0.0001	0.1069	0.6669	0.8376	0.8715	No	-4300.29
Particle Filter	-0.0000	-0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-4390.33

Exogenous spec; pivot = BTC=F. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0568 - 0.0123 x_t \quad (721)$$

$$\mu_2 = 0.1104 - 0.0051 x_t \quad (722)$$

$$\sigma_t^2(1) = 0.2432 + 0.1515y_{t-1}^2 + 0.8484\sigma_{t-1}^2 \quad (723)$$

$$\sigma_t^2(2) = 0.0019 + 0.0919y_{t-1}^2 + 0.7138\sigma_{t-1}^2 \quad (724)$$

$$P = \begin{pmatrix} 0.8589 & 0.1411 \\ 0.1455 & 0.8545 \end{pmatrix} \quad (725)$$

LogLik = -4259.78; Converged: No.

Plug-in Method

$$\mu_1 = -0.0001 - 0.0003x_t \quad (726)$$

$$\mu_2 = 0.0957 + 0.0051x_t \quad (727)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (728)$$

$$\sigma_t^2(2) = 0.2453 + 0.2590y_{t-1}^2 + 0.7410\sigma_{t-1}^2 \quad (729)$$

$$P = \begin{pmatrix} 0.5016 & 0.4984 \\ 0.2245 & 0.7755 \end{pmatrix} \quad (730)$$

LogLik = -1717.33; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0603 - 0.0230x_t \quad (731)$$

$$\mu_2 = 0.1124 + 0.0022x_t \quad (732)$$

$$\sigma_t^2(1) = 0.2318 + 0.1811y_{t-1}^2 + 0.8172\sigma_{t-1}^2 \quad (733)$$

$$\sigma_t^2(2) = 0.0001 + 0.1069y_{t-1}^2 + 0.6669\sigma_{t-1}^2 \quad (734)$$

$$P = \begin{pmatrix} 0.8376 & 0.1624 \\ 0.1285 & 0.8715 \end{pmatrix} \quad (735)$$

LogLik = -4300.29; Converged: No.

Particle Filter Method

$$\mu_1 = -0.0000 + 0.0500x_t \quad (736)$$

$$\mu_2 = -0.0000 + 0.0500x_t \quad (737)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500y_{t-1}^2 + 0.8500\sigma_{t-1}^2 \quad (738)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800y_{t-1}^2 + 0.7800\sigma_{t-1}^2 \quad (739)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (740)$$

LogLik = -4390.33; Converged: Yes.

4.3. Asset: VIX

Baseline Specification

Table 38: VIX — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.3530	0.3507	–	–	0.4207	0.1111	0.8876	0.2340	0.1312	0.7930	0.9664	0.7423	No	-9369.61
Plug-in	0.3015	-0.3016	–	–	0.4032	0.1087	0.8894	0.2142	0.1188	0.7826	0.9513	0.7525	No	-9248.85
Quadrature	0.3551	0.2797	–	–	0.4243	0.1126	0.8873	0.2380	0.1332	0.7932	0.9657	0.7407	No	-9364.19
Particle Filter	0.0000	0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-11098.09

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.3530 \quad (741)$$

$$\mu_2 = 0.3507 \quad (742)$$

$$\sigma_t^2(1) = 0.4207 + 0.1111 y_{t-1}^2 + 0.8876 \sigma_{t-1}^2 \quad (743)$$

$$\sigma_t^2(2) = 0.2340 + 0.1312 y_{t-1}^2 + 0.7930 \sigma_{t-1}^2 \quad (744)$$

$$P = \begin{pmatrix} 0.9664 & 0.0336 \\ 0.2577 & 0.7423 \end{pmatrix} \quad (745)$$

LogLik = -9369.61 ; Converged: No.

Plug-in Method

$$\mu_1 = 0.3015 \quad (746)$$

$$\mu_2 = -0.3016 \quad (747)$$

$$\sigma_t^2(1) = 0.4032 + 0.1087 y_{t-1}^2 + 0.8894 \sigma_{t-1}^2 \quad (748)$$

$$\sigma_t^2(2) = 0.2142 + 0.1188 y_{t-1}^2 + 0.7826 \sigma_{t-1}^2 \quad (749)$$

$$P = \begin{pmatrix} 0.9513 & 0.0487 \\ 0.2475 & 0.7525 \end{pmatrix} \quad (750)$$

LogLik = -9248.85 ; Converged: No.

Quadrature Method

$$\mu_1 = 0.3551 \quad (751)$$

$$\mu_2 = 0.2797 \quad (752)$$

$$\sigma_t^2(1) = 0.4243 + 0.1126y_{t-1}^2 + 0.8873\sigma_{t-1}^2 \quad (753)$$

$$\sigma_t^2(2) = 0.2380 + 0.1332y_{t-1}^2 + 0.7932\sigma_{t-1}^2 \quad (754)$$

$$P = \begin{pmatrix} 0.9657 & 0.0343 \\ 0.2593 & 0.7407 \end{pmatrix} \quad (755)$$

LogLik = -9364.19; Converged: No.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (756)$$

$$\mu_2 = 0.0000 \quad (757)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500y_{t-1}^2 + 0.8500\sigma_{t-1}^2 \quad (758)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800y_{t-1}^2 + 0.7800\sigma_{t-1}^2 \quad (759)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (760)$$

LogLik = -11098.09; Converged: Yes.

Exogenous Specification**Table 39:** VIX — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.4237	0.3489	0.4686	-0.3806	0.4444	0.1177	0.8811	0.2612	0.1498	0.7963	0.9416	0.7347	No	-9321.98
Plug-in	0.4222	-0.4223	0.4677	-0.3767	0.4437	0.1220	0.8776	0.2604	0.1473	0.7879	0.9459	0.7346	No	-9284.58
Quadrature	0.7128	0.7112	-0.7926	0.8766	0.7166	0.2768	0.7224	0.6291	0.3050	0.6538	0.7899	0.8031	No	-9503.13
Particle Filter	0.0000	0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-11097.16

Exogenous spec; pivot = BTC=F. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.4237 + 0.4686x_t \quad (761)$$

$$\mu_2 = 0.3489 - 0.3806x_t \quad (762)$$

$$\sigma_t^2(1) = 0.4444 + 0.1177 y_{t-1}^2 + 0.8811 \sigma_{t-1}^2 \quad (763)$$

$$\sigma_t^2(2) = 0.2612 + 0.1498 y_{t-1}^2 + 0.7963 \sigma_{t-1}^2 \quad (764)$$

$$P = \begin{pmatrix} 0.9416 & 0.0584 \\ 0.2653 & 0.7347 \end{pmatrix} \quad (765)$$

LogLik = -9321.98; Converged: No.

Plug-in Method

$$\mu_1 = 0.4222 + 0.4677 x_t \quad (766)$$

$$\mu_2 = -0.4223 - 0.3767 x_t \quad (767)$$

$$\sigma_t^2(1) = 0.4437 + 0.1220 y_{t-1}^2 + 0.8776 \sigma_{t-1}^2 \quad (768)$$

$$\sigma_t^2(2) = 0.2604 + 0.1473 y_{t-1}^2 + 0.7879 \sigma_{t-1}^2 \quad (769)$$

$$P = \begin{pmatrix} 0.9459 & 0.0541 \\ 0.2654 & 0.7346 \end{pmatrix} \quad (770)$$

LogLik = -9284.58; Converged: No.

Quadrature Method

$$\mu_1 = 0.7128 - 0.7926 x_t \quad (771)$$

$$\mu_2 = 0.7112 + 0.8766 x_t \quad (772)$$

$$\sigma_t^2(1) = 0.7166 + 0.2768 y_{t-1}^2 + 0.7224 \sigma_{t-1}^2 \quad (773)$$

$$\sigma_t^2(2) = 0.6291 + 0.3050 y_{t-1}^2 + 0.6538 \sigma_{t-1}^2 \quad (774)$$

$$P = \begin{pmatrix} 0.7899 & 0.2101 \\ 0.1969 & 0.8031 \end{pmatrix} \quad (775)$$

LogLik = -9503.13; Converged: No.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500 x_t \quad (776)$$

$$\mu_2 = 0.0000 + 0.0500 x_t \quad (777)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (778)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (779)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (780)$$

LogLik = -11097.16; Converged: Yes.

4.4. Asset: GC=F

Baseline Specification

Table 40: GC=F — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0482	0.0002	–	–	0.8257	0.0482	0.0001	0.0001	0.0001	0.0001	0.7494	0.5474	Yes	-1105.49
Plug-in	0.0357	0.0001	–	–	0.0416	0.0848	0.9000	0.0001	0.0001	0.0001	0.7598	0.5127	Yes	-884.73
Quadrature	0.0497	0.0002	–	–	0.8321	0.0493	0.0001	0.0001	0.0001	0.0001	0.7491	0.5464	Yes	-1130.01
Particle Filter	0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3267.32

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0482 \quad (781)$$

$$\mu_2 = 0.0002 \quad (782)$$

$$\sigma_t^2(1) = 0.8257 + 0.0482 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (783)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (784)$$

$$P = \begin{pmatrix} 0.7494 & 0.2506 \\ 0.4526 & 0.5474 \end{pmatrix} \quad (785)$$

LogLik = -1105.49; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0357 \quad (786)$$

$$\mu_2 = 0.0001 \quad (787)$$

$$\sigma_t^2(1) = 0.0416 + 0.0848 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (788)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (789)$$

$$P = \begin{pmatrix} 0.7598 & 0.2402 \\ 0.4873 & 0.5127 \end{pmatrix} \quad (790)$$

LogLik = -884.73; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0497 \quad (791)$$

$$\mu_2 = 0.0002 \quad (792)$$

$$\sigma_t^2(1) = 0.8321 + 0.0493 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (793)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (794)$$

$$P = \begin{pmatrix} 0.7491 & 0.2509 \\ 0.4536 & 0.5464 \end{pmatrix} \quad (795)$$

LogLik = -1130.01; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (796)$$

$$\mu_2 = 0.0000 \quad (797)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (798)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (799)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (800)$$

LogLik = -3267.32; Converged: Yes.

Exogenous Specification**Table 41:** GC=F — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0467	0.0002	-0.0008	0.0002	0.8268	0.0481	0.0001	0.0001	0.0001	0.0001	0.7484	0.5471	Yes	-1105.22
Plug-in	0.0356	0.0001	-0.0023	-0.0000	0.0415	0.0850	0.9000	0.0001	0.0001	0.0001	0.7598	0.5129	Yes	-884.64
Quadrature	0.0479	0.0003	-0.0010	0.0002	0.8316	0.0497	0.0001	0.0001	0.0001	0.0001	0.7491	0.5468	Yes	-1129.58
Particle Filter	0.0000	-0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-3551.41

Exogenous spec; pivot = BTC=F. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0467 - 0.0008 x_t \quad (801)$$

$$\mu_2 = 0.0002 + 0.0002 x_t \quad (802)$$

$$\sigma_t^2(1) = 0.8268 + 0.0481 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (803)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (804)$$

$$P = \begin{pmatrix} 0.7484 & 0.2516 \\ 0.4529 & 0.5471 \end{pmatrix} \quad (805)$$

LogLik = -1105.22; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0356 - 0.0023 x_t \quad (806)$$

$$\mu_2 = 0.0001 - 0.0000 x_t \quad (807)$$

$$\sigma_t^2(1) = 0.0415 + 0.0850 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (808)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (809)$$

$$P = \begin{pmatrix} 0.7598 & 0.2402 \\ 0.4871 & 0.5129 \end{pmatrix} \quad (810)$$

LogLik = -884.64; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0479 - 0.0010 x_t \quad (811)$$

$$\mu_2 = 0.0003 + 0.0002 x_t \quad (812)$$

$$\sigma_t^2(1) = 0.8316 + 0.0497 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (813)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (814)$$

$$P = \begin{pmatrix} 0.7491 & 0.2509 \\ 0.4532 & 0.5468 \end{pmatrix} \quad (815)$$

LogLik = -1129.58; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000 x_t \quad (816)$$

$$\mu_2 = -0.0000 + 0.1000 x_t \quad (817)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (818)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (819)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (820)$$

LogLik = -3551.41; Converged: Yes.

4.5. Asset: CL=F

Baseline Specification

Table 42: CL=F — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.1029	0.4738	—	—	0.2855	0.0985	0.9000	0.0001	0.1613	0.5101	0.9353	0.6840	Yes	-5807.49
Plug-in	-0.0540	0.0125	—	—	0.5635	0.1940	0.8059	0.3725	0.2065	0.6159	0.7590	0.9173	No	-5767.89
Quadrature	0.0587	0.0703	—	—	0.0846	0.0764	0.9000	0.2353	0.2164	0.7820	0.9900	0.8262	Yes	-5819.91
Particle Filter	0.0000	0.0000	—	—	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	Yes	-5874.11

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.1029 \quad (821)$$

$$\mu_2 = 0.4738 \quad (822)$$

$$\sigma_t^2(1) = 0.2855 + 0.0985 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (823)$$

$$\sigma_t^2(2) = 0.0001 + 0.1613 y_{t-1}^2 + 0.5101 \sigma_{t-1}^2 \quad (824)$$

$$P = \begin{pmatrix} 0.9353 & 0.0647 \\ 0.3160 & 0.6840 \end{pmatrix} \quad (825)$$

LogLik = -5807.49; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0540 \quad (826)$$

$$\mu_2 = 0.0125 \quad (827)$$

$$\sigma_t^2(1) = 0.5635 + 0.1940 y_{t-1}^2 + 0.8059 \sigma_{t-1}^2 \quad (828)$$

$$\sigma_t^2(2) = 0.3725 + 0.2065 y_{t-1}^2 + 0.6159 \sigma_{t-1}^2 \quad (829)$$

$$P = \begin{pmatrix} 0.7590 & 0.2410 \\ 0.0827 & 0.9173 \end{pmatrix} \quad (830)$$

LogLik = -5767.89; Converged: No.

Quadrature Method

$$\mu_1 = 0.0587 \quad (831)$$

$$\mu_2 = 0.0703 \quad (832)$$

$$\sigma_t^2(1) = 0.0846 + 0.0764 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (833)$$

$$\sigma_t^2(2) = 0.2353 + 0.2164 y_{t-1}^2 + 0.7820 \sigma_{t-1}^2 \quad (834)$$

$$P = \begin{pmatrix} 0.9900 & 0.0100 \\ 0.1738 & 0.8262 \end{pmatrix} \quad (835)$$

LogLik = -5819.91; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (836)$$

$$\mu_2 = 0.0000 \quad (837)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (838)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (839)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (840)$$

LogLik = -5874.11; Converged: Yes.

Exogenous Specification

Table 43: CL=F — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0456	0.0593	-0.0001	-0.0137	0.2848	0.1052	0.8945	0.1006	0.0789	0.7595	0.9007	0.8103	No	-5823.05
Plug-in	0.0656	0.0005	0.0761	0.0005	0.2946	0.1629	0.8371	0.0001	0.0001	0.1781	0.7478	0.4320	No	-3123.80
Quadrature	0.0781	0.1144	0.0027	-0.0241	0.1253	0.0606	0.9000	0.2606	0.2353	0.7647	0.9884	0.8334	Yes	-5823.02
Particle Filter	0.0000	0.0000	0.0500	0.0500	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	No	-5884.73

Exogenous spec; pivot = BTC=F. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0456 - 0.0001 x_t \quad (841)$$

$$\mu_2 = 0.0593 - 0.0137 x_t \quad (842)$$

$$\sigma_t^2(1) = 0.2848 + 0.1052y_{t-1}^2 + 0.8945\sigma_{t-1}^2 \quad (843)$$

$$\sigma_t^2(2) = 0.1006 + 0.0789y_{t-1}^2 + 0.7595\sigma_{t-1}^2 \quad (844)$$

$$P = \begin{pmatrix} 0.9007 & 0.0993 \\ 0.1897 & 0.8103 \end{pmatrix} \quad (845)$$

LogLik = -5823.05; Converged: No.

Plug-in Method

$$\mu_1 = 0.0656 + 0.0761x_t \quad (846)$$

$$\mu_2 = 0.0005 + 0.0005x_t \quad (847)$$

$$\sigma_t^2(1) = 0.2946 + 0.1629y_{t-1}^2 + 0.8371\sigma_{t-1}^2 \quad (848)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.1781\sigma_{t-1}^2 \quad (849)$$

$$P = \begin{pmatrix} 0.7478 & 0.2522 \\ 0.5680 & 0.4320 \end{pmatrix} \quad (850)$$

LogLik = -3123.80; Converged: No.

Quadrature Method

$$\mu_1 = 0.0781 + 0.0027x_t \quad (851)$$

$$\mu_2 = 0.1144 - 0.0241x_t \quad (852)$$

$$\sigma_t^2(1) = 0.1253 + 0.0606y_{t-1}^2 + 0.9000\sigma_{t-1}^2 \quad (853)$$

$$\sigma_t^2(2) = 0.2606 + 0.2353y_{t-1}^2 + 0.7647\sigma_{t-1}^2 \quad (854)$$

$$P = \begin{pmatrix} 0.9884 & 0.0116 \\ 0.1666 & 0.8334 \end{pmatrix} \quad (855)$$

LogLik = -5823.02; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.0500x_t \quad (856)$$

$$\mu_2 = 0.0000 + 0.0500x_t \quad (857)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500y_{t-1}^2 + 0.8500\sigma_{t-1}^2 \quad (858)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800y_{t-1}^2 + 0.7800\sigma_{t-1}^2 \quad (859)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (860)$$

LogLik = -5884.73; Converged: No.

4.6. Asset: BTC=F

Baseline Specification

Table 44: BTC=F — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.4107	-0.1779	–	–	0.6992	0.0041	0.9000	0.5715	0.1156	0.8784	0.8169	0.7175	Yes	-7441.63
Plug-in	0.7002	0.6944	–	–	0.7114	0.2689	0.7137	0.6196	0.2778	0.5905	0.7869	0.8226	No	-7692.15
Quadrature	0.4257	-0.1560	–	–	0.6956	0.0037	0.9000	0.5464	0.1097	0.8836	0.8764	0.8495	Yes	-7441.91
Particle Filter	-0.0000	0.0000	–	–	0.3000	0.0500	0.8500	0.1000	0.0800	0.7800	0.9500	0.8000	No	-8163.70

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.4107 \quad (861)$$

$$\mu_2 = -0.1779 \quad (862)$$

$$\sigma_t^2(1) = 0.6992 + 0.0041 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (863)$$

$$\sigma_t^2(2) = 0.5715 + 0.1156 y_{t-1}^2 + 0.8784 \sigma_{t-1}^2 \quad (864)$$

$$P = \begin{pmatrix} 0.8169 & 0.1831 \\ 0.2825 & 0.7175 \end{pmatrix} \quad (865)$$

LogLik = -7441.63; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.7002 \quad (866)$$

$$\mu_2 = 0.6944 \quad (867)$$

$$\sigma_t^2(1) = 0.7114 + 0.2689 y_{t-1}^2 + 0.7137 \sigma_{t-1}^2 \quad (868)$$

$$\sigma_t^2(2) = 0.6196 + 0.2778 y_{t-1}^2 + 0.5905 \sigma_{t-1}^2 \quad (869)$$

$$P = \begin{pmatrix} 0.7869 & 0.2131 \\ 0.1774 & 0.8226 \end{pmatrix} \quad (870)$$

LogLik = -7692.15; Converged: No.

Quadrature Method

$$\mu_1 = 0.4257 \quad (871)$$

$$\mu_2 = -0.1560 \quad (872)$$

$$\sigma_t^2(1) = 0.6956 + 0.0037 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (873)$$

$$\sigma_t^2(2) = 0.5464 + 0.1097 y_{t-1}^2 + 0.8836 \sigma_{t-1}^2 \quad (874)$$

$$P = \begin{pmatrix} 0.8764 & 0.1236 \\ 0.1505 & 0.8495 \end{pmatrix} \quad (875)$$

LogLik = -7441.91; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (876)$$

$$\mu_2 = 0.0000 \quad (877)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500 y_{t-1}^2 + 0.8500 \sigma_{t-1}^2 \quad (878)$$

$$\sigma_t^2(2) = 0.1000 + 0.0800 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (879)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \quad (880)$$

LogLik = -8163.70; Converged: No.

5. Foreign Exchange Market

Pivot (exogenous conditioning variable): EURUSD=X.

5.1. Asset: GBPUSD=X

Baseline Specification

Table 45: GBPUSD=X — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0002	-0.0002	–	–	0.0948	0.1729	0.8264	0.0001	0.0001	0.0001	0.7522	0.4401	Yes	-51.61
Plug-in	-0.0015	-0.0001	–	–	0.0120	0.0905	0.9000	0.0001	0.0001	0.0009	0.7643	0.4823	Yes	63.64
Quadrature	-0.0079	-0.0002	–	–	0.0001	0.1098	0.8891	0.0001	0.0001	0.0001	0.7627	0.4483	Yes	-76.25
Particle Filter	0.0000	0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-2116.49

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0002 \quad (881)$$

$$\mu_2 = -0.0002 \quad (882)$$

$$\sigma_t^2(1) = 0.0948 + 0.1729 y_{t-1}^2 + 0.8264 \sigma_{t-1}^2 \quad (883)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (884)$$

$$P = \begin{pmatrix} 0.7522 & 0.2478 \\ 0.5599 & 0.4401 \end{pmatrix} \quad (885)$$

LogLik = -51.61; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0015 \quad (886)$$

$$\mu_2 = -0.0001 \quad (887)$$

$$\sigma_t^2(1) = 0.0120 + 0.0905 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (888)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0009 \sigma_{t-1}^2 \quad (889)$$

$$P = \begin{pmatrix} 0.7643 & 0.2357 \\ 0.5177 & 0.4823 \end{pmatrix} \quad (890)$$

LogLik = 63.64; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0079 \quad (891)$$

$$\mu_2 = -0.0002 \quad (892)$$

$$\sigma_t^2(1) = 0.0001 + 0.1098 y_{t-1}^2 + 0.8891 \sigma_{t-1}^2 \quad (893)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (894)$$

$$P = \begin{pmatrix} 0.7627 & 0.2373 \\ 0.5517 & 0.4483 \end{pmatrix} \quad (895)$$

LogLik = -76.25; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (896)$$

$$\mu_2 = 0.0000 \quad (897)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (898)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (899)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (900)$$

LogLik = -2116.49; Converged: Yes.

Exogenous Specification

Table 46: GBPUSD=X — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0002	-0.0019	0.0007	0.0401	0.0001	0.0001	0.0001	0.0987	0.1496	0.8503	0.4826	0.7470	Yes	-54.16
Plug-in	-0.0015	-0.0001	0.0086	0.0005	0.0120	0.0905	0.9000	0.0001	0.0001	0.0009	0.7644	0.4822	Yes	63.75
Quadrature	0.0014	0.0001	0.0177	0.0007	0.0189	0.2002	0.7965	0.0001	0.0001	0.0001	0.7483	0.4411	Yes	-78.17
Particle Filter	0.0000	-0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-2120.49

Exogenous spec; pivot = EURUSD=X. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0002 + 0.0007 x_t \quad (901)$$

$$\mu_2 = -0.0019 + 0.0401 x_t \quad (902)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (903)$$

$$\sigma_t^2(2) = 0.0987 + 0.1496 y_{t-1}^2 + 0.8503 \sigma_{t-1}^2 \quad (904)$$

$$P = \begin{pmatrix} 0.4826 & 0.5174 \\ 0.2530 & 0.7470 \end{pmatrix} \quad (905)$$

LogLik = -54.16; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0015 + 0.0086 x_t \quad (906)$$

$$\mu_2 = -0.0001 + 0.0005 x_t \quad (907)$$

$$\sigma_t^2(1) = 0.0120 + 0.0905 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (908)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0009 \sigma_{t-1}^2 \quad (909)$$

$$P = \begin{pmatrix} 0.7644 & 0.2356 \\ 0.5178 & 0.4822 \end{pmatrix} \quad (910)$$

LogLik = 63.75; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0014 + 0.0177 x_t \quad (911)$$

$$\mu_2 = 0.0001 + 0.0007 x_t \quad (912)$$

$$\sigma_t^2(1) = 0.0189 + 0.2002 y_{t-1}^2 + 0.7965 \sigma_{t-1}^2 \quad (913)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (914)$$

$$P = \begin{pmatrix} 0.7483 & 0.2517 \\ 0.5589 & 0.4411 \end{pmatrix} \quad (915)$$

LogLik = -78.17; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000 x_t \quad (916)$$

$$\mu_2 = -0.0000 + 0.1000 x_t \quad (917)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (918)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (919)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (920)$$

LogLik = -2120.49; Converged: Yes.

5.2. Asset: JPY=X

Baseline Specification

Table 47: JPY=X — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0452	0.0000	—	—	0.0919	0.1881	0.8119	0.0001	0.0001	0.0001	0.7805	0.4230	No	-64.72
Plug-in	-0.0002	0.0195	—	—	0.0001	0.0001	0.0001	0.0362	0.3056	0.6944	0.4976	0.7621	No	71.36
Quadrature	0.0234	0.0004	—	—	0.0010	0.2142	0.7854	0.0001	0.0001	0.0001	0.7634	0.3640	Yes	-78.14
Particle Filter	0.0000	0.0000	—	—	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-2108.91

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0452 \quad (921)$$

$$\mu_2 = 0.0000 \quad (922)$$

$$\sigma_t^2(1) = 0.0919 + 0.1881 y_{t-1}^2 + 0.8119 \sigma_{t-1}^2 \quad (923)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (924)$$

$$P = \begin{pmatrix} 0.7805 & 0.2195 \\ 0.5770 & 0.4230 \end{pmatrix} \quad (925)$$

LogLik = -64.72; Converged: No.

Plug-in Method

$$\mu_1 = -0.0002 \quad (926)$$

$$\mu_2 = 0.0195 \quad (927)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (928)$$

$$\sigma_t^2(2) = 0.0362 + 0.3056 y_{t-1}^2 + 0.6944 \sigma_{t-1}^2 \quad (929)$$

$$P = \begin{pmatrix} 0.4976 & 0.5024 \\ 0.2379 & 0.7621 \end{pmatrix} \quad (930)$$

LogLik = 71.36; Converged: No.

Quadrature Method

$$\mu_1 = 0.0234 \quad (931)$$

$$\mu_2 = 0.0004 \quad (932)$$

$$\sigma_t^2(1) = 0.0010 + 0.2142 y_{t-1}^2 + 0.7854 \sigma_{t-1}^2 \quad (933)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (934)$$

$$P = \begin{pmatrix} 0.7634 & 0.2366 \\ 0.6360 & 0.3640 \end{pmatrix} \quad (935)$$

LogLik = -78.14; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (936)$$

$$\mu_2 = 0.0000 \quad (937)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (938)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (939)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (940)$$

LogLik = -2108.91; Converged: No.

Exogenous Specification

Table 48: JPY=X — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0001	0.0309	0.0025	-0.0809	0.0001	0.0001	0.0001	0.0875	0.2330	0.7670	0.4197	0.7736	No	-66.56
Plug-in	0.0001	0.0682	0.0024	-0.0378	0.0001	0.0001	0.0001	0.0358	0.3058	0.6942	0.4986	0.7820	No	63.15
Quadrature	0.0000	0.0095	0.0014	-0.1335	0.0001	0.0001	0.0001	0.0085	0.1774	0.8223	0.4331	0.7419	No	-74.43
Particle Filter	0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-2116.35

Exogenous spec; pivot = EURUSD=X. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0001 + 0.0025 x_t \quad (941)$$

$$\mu_2 = 0.0309 - 0.0809 x_t \quad (942)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (943)$$

$$\sigma_t^2(2) = 0.0875 + 0.2330 y_{t-1}^2 + 0.7670 \sigma_{t-1}^2 \quad (944)$$

$$P = \begin{pmatrix} 0.4197 & 0.5803 \\ 0.2264 & 0.7736 \end{pmatrix} \quad (945)$$

LogLik = -66.56; Converged: No.

Plug-in Method

$$\mu_1 = 0.0001 + 0.0024 x_t \quad (946)$$

$$\mu_2 = 0.0682 - 0.0378 x_t \quad (947)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (948)$$

$$\sigma_t^2(2) = 0.0358 + 0.3058 y_{t-1}^2 + 0.6942 \sigma_{t-1}^2 \quad (949)$$

$$P = \begin{pmatrix} 0.4986 & 0.5014 \\ 0.2180 & 0.7820 \end{pmatrix} \quad (950)$$

LogLik = 63.15; Converged: No.

Quadrature Method

$$\mu_1 = 0.0000 + 0.0014 x_t \quad (951)$$

$$\mu_2 = 0.0095 - 0.1335 x_t \quad (952)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (953)$$

$$\sigma_t^2(2) = 0.0085 + 0.1774 y_{t-1}^2 + 0.8223 \sigma_{t-1}^2 \quad (954)$$

$$P = \begin{pmatrix} 0.4331 & 0.5669 \\ 0.2581 & 0.7419 \end{pmatrix} \quad (955)$$

LogLik = -74.43; Converged: No.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000 x_t \quad (956)$$

$$\mu_2 = 0.0000 + 0.1000 x_t \quad (957)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (958)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (959)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (960)$$

LogLik = -2116.35; Converged: No.

5.3. Asset: AUDUSD=X

Baseline Specification

Table 49: AUDUSD=X — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0004	0.0125	—	—	0.0001	0.0001	0.0001	0.1190	0.1000	0.9000	0.4328	0.7540	Yes	-510.71
Plug-in	-0.0000	-0.0158	—	—	0.0001	0.0001	0.0001	0.0174	0.0866	0.9000	0.4713	0.7732	Yes	-376.14
Quadrature	-0.0126	0.0003	—	—	0.0350	0.1698	0.8276	0.0001	0.0001	0.0001	0.7515	0.4062	Yes	-540.09
Particle Filter	0.0000	-0.0000	—	—	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-2384.22

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.0004 \quad (961)$$

$$\mu_2 = 0.0125 \quad (962)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (963)$$

$$\sigma_t^2(2) = 0.1190 + 0.1000 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (964)$$

$$P = \begin{pmatrix} 0.4328 & 0.5672 \\ 0.2460 & 0.7540 \end{pmatrix} \quad (965)$$

LogLik = -510.71; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0000 \quad (966)$$

$$\mu_2 = -0.0158 \quad (967)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (968)$$

$$\sigma_t^2(2) = 0.0174 + 0.0866 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (969)$$

$$P = \begin{pmatrix} 0.4713 & 0.5287 \\ 0.2268 & 0.7732 \end{pmatrix} \quad (970)$$

LogLik = -376.14; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0126 \quad (971)$$

$$\mu_2 = 0.0003 \quad (972)$$

$$\sigma_t^2(1) = 0.0350 + 0.1698 y_{t-1}^2 + 0.8276 \sigma_{t-1}^2 \quad (973)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (974)$$

$$P = \begin{pmatrix} 0.7515 & 0.2485 \\ 0.5938 & 0.4062 \end{pmatrix} \quad (975)$$

LogLik = -540.09; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (976)$$

$$\mu_2 = -0.0000 \quad (977)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (978)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (979)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (980)$$

LogLik = -2384.22; Converged: Yes.

Exogenous Specification

Table 50: AUDUSD=X — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0068	0.0002	-0.0928	-0.0012	0.1200	0.0997	0.9000	0.0001	0.0001	0.0001	0.7479	0.4420	Yes	-509.76
Plug-in	-0.0163	-0.0000	-0.0526	-0.0007	0.0173	0.0869	0.9000	0.0001	0.0001	0.0001	0.7731	0.4712	Yes	-375.10
Quadrature	-0.0015	0.0000	-0.0135	-0.0012	0.0226	0.1004	0.8987	0.0001	0.0001	0.0001	0.7533	0.4427	Yes	-532.90
Particle Filter	0.0000	-0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-2390.26

Exogenous spec; pivot = EURUSD=X. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0068 - 0.0928 x_t \quad (981)$$

$$\mu_2 = 0.0002 - 0.0012 x_t \quad (982)$$

$$\sigma_t^2(1) = 0.1200 + 0.0997 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (983)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (984)$$

$$P = \begin{pmatrix} 0.7479 & 0.2521 \\ 0.5580 & 0.4420 \end{pmatrix} \quad (985)$$

LogLik = -509.76; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0163 - 0.0526 x_t \quad (986)$$

$$\mu_2 = -0.0000 - 0.0007 x_t \quad (987)$$

$$\sigma_t^2(1) = 0.0173 + 0.0869 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (988)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (989)$$

$$P = \begin{pmatrix} 0.7731 & 0.2269 \\ 0.5288 & 0.4712 \end{pmatrix} \quad (990)$$

LogLik = -375.10; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0015 - 0.0135 x_t \quad (991)$$

$$\mu_2 = 0.0000 - 0.0012 x_t \quad (992)$$

$$\sigma_t^2(1) = 0.0226 + 0.1004 y_{t-1}^2 + 0.8987 \sigma_{t-1}^2 \quad (993)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (994)$$

$$P = \begin{pmatrix} 0.7533 & 0.2467 \\ 0.5573 & 0.4427 \end{pmatrix} \quad (995)$$

LogLik = -532.90; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000 x_t \quad (996)$$

$$\mu_2 = -0.0000 + 0.1000 x_t \quad (997)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (998)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (999)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (1000)$$

LogLik = -2390.26; Converged: Yes.

5.4. Asset: CHF=X

Baseline Specification

Table 51: CHF=X — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0094	0.0003	—	—	0.0687	0.2163	0.7800	0.0001	0.0001	0.0001	0.7638	0.4721	Yes	302.30
Plug-in	-0.0005	0.0002	—	—	0.0244	0.1622	0.7876	0.0001	0.0001	0.0001	0.7655	0.4830	Yes	372.12
Quadrature	0.0008	0.0002	—	—	0.0001	0.1670	0.8114	0.0001	0.0001	0.0001	0.7539	0.4700	Yes	286.83
Particle Filter	0.0000	0.0000	—	—	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	No	-1913.41

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0094 \quad (1001)$$

$$\mu_2 = 0.0003 \quad (1002)$$

$$\sigma_t^2(1) = 0.0687 + 0.2163 y_{t-1}^2 + 0.7800 \sigma_{t-1}^2 \quad (1003)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1004)$$

$$P = \begin{pmatrix} 0.7638 & 0.2362 \\ 0.5279 & 0.4721 \end{pmatrix} \quad (1005)$$

LogLik = 302.30; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0005 \quad (1006)$$

$$\mu_2 = 0.0002 \quad (1007)$$

$$\sigma_t^2(1) = 0.0244 + 0.1622 y_{t-1}^2 + 0.7876 \sigma_{t-1}^2 \quad (1008)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1009)$$

$$P = \begin{pmatrix} 0.7655 & 0.2345 \\ 0.5170 & 0.4830 \end{pmatrix} \quad (1010)$$

LogLik = 372.12; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0008 \quad (1011)$$

$$\mu_2 = 0.0002 \quad (1012)$$

$$\sigma_t^2(1) = 0.0001 + 0.1670 y_{t-1}^2 + 0.8114 \sigma_{t-1}^2 \quad (1013)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1014)$$

$$P = \begin{pmatrix} 0.7539 & 0.2461 \\ 0.5300 & 0.4700 \end{pmatrix} \quad (1015)$$

LogLik = 286.83; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 \quad (1016)$$

$$\mu_2 = 0.0000 \quad (1017)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (1018)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (1019)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (1020)$$

LogLik = -1913.41; Converged: No.

Exogenous Specification

Table 52: CHF=X — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0039	0.0000	0.0172	0.0007	0.0715	0.2024	0.7974	0.0001	0.0001	0.0001	0.7560	0.4638	Yes	303.39
Plug-in	0.0002	-0.0005	0.0004	-0.0164	0.0001	0.0001	0.0001	0.0244	0.1615	0.7882	0.4831	0.7654	Yes	372.35
Quadrature	0.0002	0.0006	0.0006	-0.0223	0.0001	0.0001	0.0001	0.0001	0.1667	0.8117	0.4697	0.7537	Yes	287.22
Particle Filter	-0.0000	0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-1921.41

Exogenous spec; pivot = EURUSD=X. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = -0.0039 + 0.0172 x_t \quad (1021)$$

$$\mu_2 = 0.0000 + 0.0007 x_t \quad (1022)$$

$$\sigma_t^2(1) = 0.0715 + 0.2024y_{t-1}^2 + 0.7974\sigma_{t-1}^2 \quad (1023)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (1024)$$

$$P = \begin{pmatrix} 0.7560 & 0.2440 \\ 0.5362 & 0.4638 \end{pmatrix} \quad (1025)$$

LogLik = 303.39; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0002 + 0.0004x_t \quad (1026)$$

$$\mu_2 = -0.0005 - 0.0164x_t \quad (1027)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (1028)$$

$$\sigma_t^2(2) = 0.0244 + 0.1615y_{t-1}^2 + 0.7882\sigma_{t-1}^2 \quad (1029)$$

$$P = \begin{pmatrix} 0.4831 & 0.5169 \\ 0.2346 & 0.7654 \end{pmatrix} \quad (1030)$$

LogLik = 372.35; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0002 + 0.0006x_t \quad (1031)$$

$$\mu_2 = 0.0006 - 0.0223x_t \quad (1032)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001y_{t-1}^2 + 0.0001\sigma_{t-1}^2 \quad (1033)$$

$$\sigma_t^2(2) = 0.0001 + 0.1667y_{t-1}^2 + 0.8117\sigma_{t-1}^2 \quad (1034)$$

$$P = \begin{pmatrix} 0.4697 & 0.5303 \\ 0.2463 & 0.7537 \end{pmatrix} \quad (1035)$$

LogLik = 287.22; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 + 0.1000x_t \quad (1036)$$

$$\mu_2 = 0.0000 + 0.1000x_t \quad (1037)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000y_{t-1}^2 + 0.7000\sigma_{t-1}^2 \quad (1038)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000y_{t-1}^2 + 0.6000\sigma_{t-1}^2 \quad (1039)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (1040)$$

LogLik = −1921.41; Converged: Yes.

5.5. Asset: EURGBP=X

Baseline Specification

Table 53: EURGBP=X — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0046	0.0003	—	—	0.2237	0.0801	0.0036	0.0001	0.0001	0.0001	0.7494	0.4941	Yes	274.11
Plug-in	0.0114	0.0002	—	—	0.0057	0.1000	0.9000	0.0001	0.0001	0.0001	0.8046	0.6121	Yes	557.11
Quadrature	0.0003	0.0044	—	—	0.0001	0.0001	0.0001	0.2239	0.0797	0.0001	0.4959	0.7477	Yes	252.26
Particle Filter	-0.0000	0.0000	—	—	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-1878.74

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = 0.0046 \quad (1041)$$

$$\mu_2 = 0.0003 \quad (1042)$$

$$\sigma_t^2(1) = 0.2237 + 0.0801 y_{t-1}^2 + 0.0036 \sigma_{t-1}^2 \quad (1043)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1044)$$

$$P = \begin{pmatrix} 0.7494 & 0.2506 \\ 0.5059 & 0.4941 \end{pmatrix} \quad (1045)$$

LogLik = 274.11; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0114 \quad (1046)$$

$$\mu_2 = 0.0002 \quad (1047)$$

$$\sigma_t^2(1) = 0.0057 + 0.1000 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (1048)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1049)$$

$$P = \begin{pmatrix} 0.8046 & 0.1954 \\ 0.3879 & 0.6121 \end{pmatrix} \quad (1050)$$

LogLik = 557.11; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0003 \quad (1051)$$

$$\mu_2 = 0.0044 \quad (1052)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1053)$$

$$\sigma_t^2(2) = 0.2239 + 0.0797 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1054)$$

$$P = \begin{pmatrix} 0.4959 & 0.5041 \\ 0.2523 & 0.7477 \end{pmatrix} \quad (1055)$$

LogLik = 252.26; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (1056)$$

$$\mu_2 = 0.0000 \quad (1057)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (1058)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (1059)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (1060)$$

LogLik = -1878.74; Converged: Yes.

Exogenous Specification**Table 54:** EURGBP=X — Exogenous Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	0.0045	0.0003	-0.0208	-0.0002	0.2235	0.0805	0.0035	0.0001	0.0001	0.0001	0.7494	0.4944	Yes	274.41
Plug-in	0.0003	-0.0408	-0.0009	0.0103	0.0001	0.0001	0.0001	0.0190	0.2387	0.7612	0.3547	0.7669	Yes	498.56
Quadrature	0.0046	0.0003	-0.0210	-0.0002	0.2237	0.0797	0.0001	0.0001	0.0001	0.0001	0.7479	0.4958	Yes	252.55
Particle Filter	0.0000	-0.0000	0.1000	0.1000	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-1883.37

Exogenous spec; pivot = EURUSD=X. Conv. = convergence status.

Fitted Equations (Exogenous):**Gray's Method**

$$\mu_1 = 0.0045 - 0.0208 x_t \quad (1061)$$

$$\mu_2 = 0.0003 - 0.0002 x_t \quad (1062)$$

$$\sigma_t^2(1) = 0.2235 + 0.0805 y_{t-1}^2 + 0.0035 \sigma_{t-1}^2 \quad (1063)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1064)$$

$$P = \begin{pmatrix} 0.7494 & 0.2506 \\ 0.5056 & 0.4944 \end{pmatrix} \quad (1065)$$

LogLik = 274.41; Converged: Yes.

Plug-in Method

$$\mu_1 = 0.0003 - 0.0009 x_t \quad (1066)$$

$$\mu_2 = -0.0408 + 0.0103 x_t \quad (1067)$$

$$\sigma_t^2(1) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1068)$$

$$\sigma_t^2(2) = 0.0190 + 0.2387 y_{t-1}^2 + 0.7612 \sigma_{t-1}^2 \quad (1069)$$

$$P = \begin{pmatrix} 0.3547 & 0.6453 \\ 0.2331 & 0.7669 \end{pmatrix} \quad (1070)$$

LogLik = 498.56; Converged: Yes.

Quadrature Method

$$\mu_1 = 0.0046 - 0.0210 x_t \quad (1071)$$

$$\mu_2 = 0.0003 - 0.0002 x_t \quad (1072)$$

$$\sigma_t^2(1) = 0.2237 + 0.0797 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1073)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1074)$$

$$P = \begin{pmatrix} 0.7479 & 0.2521 \\ 0.5042 & 0.4958 \end{pmatrix} \quad (1075)$$

LogLik = 252.55; Converged: Yes.

Particle Filter Method

$$\mu_1 = 0.0000 + 0.1000 x_t \quad (1076)$$

$$\mu_2 = -0.0000 + 0.1000 x_t \quad (1077)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (1078)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (1079)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (1080)$$

LogLik = -1883.37; Converged: Yes.

5.6. Asset: EURUSD=X

Baseline Specification

Table 55: EURUSD=X — Baseline Specification

Method	c_1	c_2	γ_1	γ_2	ω_1	α_1	β_1	ω_2	α_2	β_2	p_{11}	p_{22}	Conv.	LogLik
Gray's	-0.0041	-0.0005	–	–	0.0697	0.0991	0.9000	0.0001	0.0001	0.0001	0.7488	0.4934	Yes	308.28
Plug-in	-0.0085	-0.0001	–	–	0.0086	0.0902	0.9000	0.0001	0.0001	0.0001	0.7669	0.4815	Yes	399.99
Quadrature	-0.0044	-0.0001	–	–	0.0001	0.1158	0.8263	0.0001	0.0001	0.0001	0.7519	0.4557	Yes	291.55
Particle Filter	-0.0000	-0.0000	–	–	0.1000	0.1000	0.7000	0.2000	0.2000	0.6000	0.9000	0.9000	Yes	-1893.04

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

Fitted Equations (Baseline):

Gray's Method

$$\mu_1 = -0.0041 \quad (1081)$$

$$\mu_2 = -0.0005 \quad (1082)$$

$$\sigma_t^2(1) = 0.0697 + 0.0991 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (1083)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1084)$$

$$P = \begin{pmatrix} 0.7488 & 0.2512 \\ 0.5066 & 0.4934 \end{pmatrix} \quad (1085)$$

LogLik = 308.28; Converged: Yes.

Plug-in Method

$$\mu_1 = -0.0085 \quad (1086)$$

$$\mu_2 = -0.0001 \quad (1087)$$

$$\sigma_t^2(1) = 0.0086 + 0.0902 y_{t-1}^2 + 0.9000 \sigma_{t-1}^2 \quad (1088)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1089)$$

$$P = \begin{pmatrix} 0.7669 & 0.2331 \\ 0.5185 & 0.4815 \end{pmatrix} \quad (1090)$$

LogLik = 399.99; Converged: Yes.

Quadrature Method

$$\mu_1 = -0.0044 \quad (1091)$$

$$\mu_2 = -0.0001 \quad (1092)$$

$$\sigma_t^2(1) = 0.0001 + 0.1158 y_{t-1}^2 + 0.8263 \sigma_{t-1}^2 \quad (1093)$$

$$\sigma_t^2(2) = 0.0001 + 0.0001 y_{t-1}^2 + 0.0001 \sigma_{t-1}^2 \quad (1094)$$

$$P = \begin{pmatrix} 0.7519 & 0.2481 \\ 0.5443 & 0.4557 \end{pmatrix} \quad (1095)$$

LogLik = 291.55; Converged: Yes.

Particle Filter Method

$$\mu_1 = -0.0000 \quad (1096)$$

$$\mu_2 = -0.0000 \quad (1097)$$

$$\sigma_t^2(1) = 0.1000 + 0.1000 y_{t-1}^2 + 0.7000 \sigma_{t-1}^2 \quad (1098)$$

$$\sigma_t^2(2) = 0.2000 + 0.2000 y_{t-1}^2 + 0.6000 \sigma_{t-1}^2 \quad (1099)$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \quad (1100)$$

LogLik = -1893.04; Converged: Yes.